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Event-Based Monocular Depth Estimation for Robotic 3D Scene Reconstruction

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Abstract

Event cameras provide high temporal resolution, low latency, high dynamic range, and minimal motion blur. These properties make them well suited for robotic perception and three-dimensional reconstruction in scenarios involving fast camera motion or challenging lighting conditions. However, their fundamentally different output data format, compared to RGB images, limits the direct application of conventional frame-based reconstruction methods.

Early event-based depth estimation methods generally produced semi-dense depth-maps concentrated around scene edges[9]. More recently, learning-based approaches have enabled denser predictions from event data, as demonstrated by Learning Monocular Dense Depth from Events [5]. Building on this development, this thesis investigates whether established learning-based multi-view stereo (MVS) techniques originally developed for RGB images can be transferred to event representations. We adapt a learning-based MVS network that is able to process spatiotemporal event voxel grids and predict dense depth maps of the observed scene. These can later be integrated to obtain dense three-dimensional reconstructions.

This work contributes a synchronized, object-centric dataset containing event streams, camera poses, and ground-truth depth measurements, as well as an experimental evaluation of an RGB-style learned feature-warping and cost-volume MVS pipeline on event data. The results demonstrate that, within the object-centric robotic setting represented by this dataset, established learning-based MVS techniques can produce accurate depth estimates from event representations and support dense three-dimensional reconstruction through volumetric fusion.

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List of Abbreviations

Abbreviation	Meaning
AbsRel	Absolute relative error
EMA	Exponential moving average
EMVS	Event-Based Multi-View Stereo
FPS	Frames per second
MAE	Mean absolute error
MVS	Multi-view stereo
ReLU	Rectified linear unit
RMSE	Root mean squared error
SDF	Signed Distance Function
TSDF	Truncated Signed Distance Function

Chapter 1

Introduction

Three-dimensional perception is a fundamental capability for autonomous and robotic systems. A geometric representation of the environment enables applications such as robotic manipulation, navigation, scene understanding, and object modelling. One common method used to generate such three-dimensional representations is estimating depth from camera observations acquired at different viewpoints and subsequently combining the estimates into a consistent three-dimensional model. Conventional approaches commonly rely on frame-based cameras, which capture complete intensity images at fixed time intervals.

Event cameras provide an alternative sensing principle. Instead of recording complete images at a fixed frame rate, each pixel independently reports changes in logarithmic brightness. These properties offer advantages over conventional cameras under rapid motion and challenging illumination conditions, making them promising for many robotic perception tasks.

However, their unconventional output also introduces significant challenges. Many established computer-vision methods cannot be applied directly to event data. The event stream must either be processed asynchronously or transformed into a structured representation, such as an event frame, time surface, or spatiotemporal voxel grid.

Recovering depth from event data is particularly challenging. Individual events do not contain absolute intensity or depth information, and isolated event observations provide insufficient information to determine scene geometry. Depth must instead be inferred from temporal patterns, learned appearance cues, or geometric correspondences across observations.

Monocular event-based depth-estimation methods can be distinguished by how they handle camera motion. Some approaches predict depth without explicitly modelling camera poses [5], while others jointly estimate depth and camera ego-motion [12]. A third category uses externally provided or previously estimated camera poses to exploit geometric consistency across observations [9, 6]. The robotic setup considered in this thesis directly provides these

camera poses. The event camera is mounted on a calibrated robotic manipulator, making the transformation between the robot and the camera coordinate system known. Consequently, camera motion does not need to be inferred from the event data. Instead, the known poses can be used to establish geometric relationships between event observations acquired from multiple viewpoints. This creates an opportunity to apply multi-view stereo (MVS) techniques, originally developed for conventional images, to event-camera data.

MVS estimates scene depth by identifying geometrically consistent observations across several calibrated views. Modern learning-based multi-view stereo methods extract image features, warp them according to candidate depth hypotheses and relative camera poses, and aggregate the resulting correspondences in a three-dimensional cost volume. The cost volume is subsequently regularized by a neural network to infer the most likely depth for each reference-view pixel. Architectures such as MVSNet [11] and CasMVSNet [4] have demonstrated that this combination of explicit projective geometry and learned feature matching can produce accurate dense depth estimates from RGB images.

Whether this formulation transfers effectively to event data is not immediately clear. Event voxel grids, generated from the asynchronous event stream, differ substantially from RGB images in their statistics, sparsity, and dependence on motion. Nevertheless, they preserve spatial and temporal information in an image-like tensor representation and can therefore be processed using convolutional feature extractors. Furthermore, these features can be transformed into a reference frame, which makes the construction of a multi-view cost volume possible. This suggests that pose-aware learning-based multi-view stereo may provide a suitable framework for combining event observations across viewpoints.

The central objective of this thesis is to investigate whether event data and camera poses provided by a robotic manipulator can be used within an MVS-based architecture, using elements that were originally designed for RGB frame input. We focus exclusively on a monocular setup in which a single camera is moved around a static object to acquire observations from multiple viewpoints, rather than using a synchronized stereo-camera pair. We adapt established MVS techniques to voxelized event data, compare the resulting model with a simpler U-Net baseline, and investigate whether its predicted depth maps can be fused to generate accurate three-dimensional reconstructions.

Chapter 2

Background

This chapter introduces the concepts most relevant to the proposed approach. It covers event cameras and event representations, learning-based multi-view stereo, and volumetric fusion.

2.1 Event Cameras

Conventional cameras capture complete intensity images at fixed time intervals. During each exposure, all pixels are measured regardless of whether the observed scene has changed. Event cameras use a fundamentally different sensing principle. Instead of recording complete frames, every pixel independently detects changes in brightness and generates data only when the change exceeds a predefined threshold [2].

Event cameras typically operate on logarithmic intensity. In the following, $L(\mathbf{u}, t)$ therefore denotes the logarithmic intensity measured at pixel $\mathbf{u} = (x, y)^\top$ at time t .

An event is triggered when the change in logarithmic brightness since the previous event at the same pixel reaches a contrast threshold C . Let t^{prev} denote the timestamp of the previous event at pixel $\mathbf{u}$. The event-generation model is then

$$pC = L(\mathbf{u}, t) - L(\mathbf{u}, t^{\text{prev}}), \quad (2.1)$$

where t is the current event timestamp and $p \in \{-1, +1\}$ is its polarity. A positive polarity indicates an increase in brightness, while a negative polarity indicates a decrease. Each event is therefore represented as

$$e = (\mathbf{u}, t, p). \quad (2.2)$$

The camera output is an asynchronous and variable-length stream of such events rather than a sequence of regularly sampled images.

Events are commonly generated when structures move across the image plane. For example, when a dark object moves over a bright background, pixels at its leading boundary observe a decrease in brightness and generate negative events. Pixels at the trailing boundary observe an increase and generate positive events. Object contours and textured regions therefore produce many events during motion, whereas uniform regions may produce few or no events. Figure 2.1 illustrates this example by comparing a conventional RGB image with the corresponding event-camera output under camera motion.

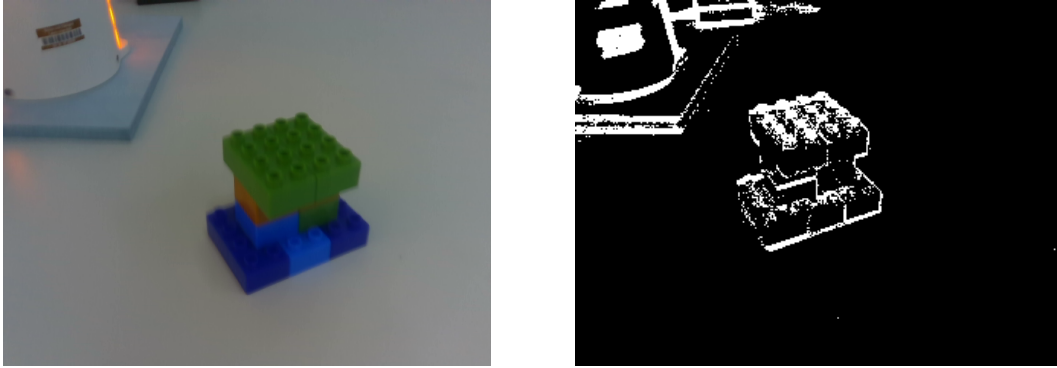


Figure 2.1: Comparison of a conventional RGB image (left) and the corresponding event-camera output (right) while under camera motion. The two camera views are close, but do not perfectly match. The events were accumulated over the duration of one RGB camera frame. For simplicity both, negative and positive events are shown as white pixels.

The number and distribution of events depend on several factors, including scene texture, illumination, camera motion, object motion, and the selected contrast threshold. A static camera observing a static scene under constant illumination produces very few events. In contrast, rapid camera motion in a textured scene can produce a large number of events within a short time interval. Event data therefore describes changes in visual appearance rather than the absolute appearance of the scene.

This sensing principle provides several advantages. Since events are generated independently at each pixel, their timestamps can provide very high temporal resolution and low latency. Furthermore, event cameras do not integrate brightness over a complete frame exposure and therefore exhibit considerably less motion blur than conventional cameras. Their logarithmic response also provides a high dynamic range, allowing them to operate in scenes containing both bright and dark regions [2]. These properties make event cameras particularly suitable for robotic applications involving fast motion or difficult illumination conditions.

However, event cameras also introduce important limitations. Individual events do not contain absolute intensity, colour, or depth information. Regions

without sufficient brightness variation may remain unobserved even though they contain valid scene geometry. Event streams can additionally contain noise caused by sensor electronics or small illumination fluctuations. Sudden global illumination changes can also trigger many events that are not caused by geometric motion.

2.2 Event Representations

The native output of an event camera is an asynchronous sequence with a variable number of events. This format is not directly compatible with conventional convolutional neural networks, which expect fixed-size tensors arranged on a regular grid. Event-based learning methods therefore commonly transform a temporal subset of the event stream into a structured representation.

A simple representation is the event frame. Events within a selected time interval are accumulated at their pixel locations. Positive and negative events may either be stored in separate channels or combined using their polarity. Event frames are compact and can be processed like conventional images. However, they discard the temporal ordering of events inside the interval. For example, an event frame cannot distinguish whether an event occurred near the beginning or near the end of the interval, as only the accumulated events are retained.

Another common representation is the time surface. A time surface stores the timestamp of the most recent event at each pixel, often transformed using an exponential decay. Recent events receive high values, while older events receive lower values. This preserves information about event recency but usually discards repeated events at the same location.

Some methods process events directly in their asynchronous form. This preserves the exact timestamps and avoids introducing an artificial temporal discretization. However, raw-event processing generally requires specialized architectures. It is less suitable for directly adapting networks originally designed for regular image tensors.

Spatiotemporal voxel grids are the representation used in this thesis. A voxel grid provides a compromise between a fully accumulated event frame and direct processing of the raw event stream. It preserves the spatial arrangement of the sensor while retaining coarse temporal information across several channels [12].

To construct a voxel grid, a temporal interval $[t_{\text{start}}, t_{\text{end}}]$ is divided into B temporal bins. For each event in this interval e_i , the timestamp is normalized to the range of bin indices:

$$\tilde{t}_i = \frac{t_i - t_{\text{start}}}{t_{\text{end}} - t_{\text{start}}}(B - 1). \quad (2.3)$$

The event is then inserted at its spatial position (x_i, y_i) . Instead of assigning it only to the nearest temporal bin, it can be distributed linearly between the two neighbouring bins:

$$V(b, x, y) = \sum_i p_i \max(0, 1 - |b - \tilde{t}_i|) \mathbb{I}[(x, y) = (x_i, y_i)], \quad (2.4)$$

where $b \in \{0, \dots, B - 1\}$ and $\mathbb{I}$ denotes the indicator function. An event positioned exactly at the temporal centre of one bin contributes fully to that bin. An event occurring between two temporal bin centres contributes to both according to its temporal distance.

The resulting voxel grid has the shape

$$V \in \mathbb{R}^{B \times H \times W}, \quad (2.5)$$

where H and W are the spatial image dimensions.

A voxel grid is not equivalent to a sequence of conventional intensity images. Its values describe accumulated brightness changes, not absolute brightness. A value of zero does not indicate a black image region. It indicates that no event contribution was recorded at the corresponding pixel and temporal bin, or that positive and negative contributions cancelled each other.

Compared with a single event frame, voxel grids preserve coarse temporal ordering. For example, the network can distinguish whether positive activity occurred near the beginning or the end of the selected interval. This can provide information about the direction and evolution of local motion. At the same time, voxel grids remain regular image-aligned tensors and can therefore be processed using standard convolutional feature extractors.

The main design parameters of a voxel representation are the duration of the temporal window and the number of temporal bins. A longer window usually contains more events and therefore produces a denser representation. However, large camera motion within the interval may cause event structures from different poses to overlap. A shorter window provides more precise temporal localization but may contain too few events, especially in uniform regions or during slow motion.

Similarly, increasing the number of bins preserves finer temporal information but produces more input channels and potentially sparser individual bins. Using too few bins merges events that occurred at substantially different times. The appropriate choice therefore depends on the event rate, camera velocity, temporal-window duration, and computational constraints.

2.3 U-Net Architectures

U-Net is a convolutional encoder–decoder architecture originally introduced for image segmentation [10]. Its structure consists of a contracting encoder path and an expanding decoder path. The encoder progressively reduces the spatial resolution of the input while increasing the number of feature channels, allowing the network to capture increasingly abstract and spatially broader information. Each encoder stage typically consists of several convolutional layers followed by ReLU activation functions, with spatial downsampling performed using a convolution with stride 2 or max pooling. The resulting feature representation is then passed to the next encoder stage.

The decoder subsequently restores the spatial resolution to produce a dense pixel-wise output. Each decoder stage first increases the spatial resolution of the feature representation, after which it is combined with features from the corresponding encoder stage through a skip connection. The combined representation is then processed by several convolutional layers followed by ReLU activation functions. Depending on the specific implementation, the upsampling operation can, for example, be implemented through bilinear upsampling.

A defining component of U-Net is the use of skip connections between corresponding encoder and decoder stages. Features from the encoder are transferred directly to decoder stages operating at the same spatial resolution. This allows high-resolution spatial information that may be lost during downsampling to be combined with the more abstract features produced at deeper layers. The decoder can therefore recover fine spatial structure while still using broader contextual information.

Although originally proposed for semantic segmentation, the general U-Net structure is also suitable for other dense prediction tasks in which an output value is required for every input pixel. In this thesis, the term U-Net is used to refer to encoder–decoder architectures that follow the characteristic U-Net structure with skip connections between corresponding encoder and decoder stages, rather than to the exact original architecture proposed by Ronneberger et al. [10].

2.4 Learning-Based Multi-View Stereo

Multi-view stereo (MVS) aims to recover the three-dimensional structure of a static scene from multiple images captured from different viewpoints. In many MVS approaches, this is achieved by estimating a dense depth-map for a selected reference image. The cameras are assumed to be calibrated, meaning that their intrinsic parameters and relative poses are known. One image is selected as the reference view, while the remaining images are treated as source views. The objective is to estimate a depth value for each pixel in the reference image.

The central principle of MVS is that a three-dimensional scene point should be observed consistently from different viewpoints. A pixel in the reference image defines a ray originating at the reference camera centre. Each possible depth along this ray corresponds to a different three-dimensional point. Using the known camera poses and intrinsics, this hypothetical point can be projected into every source image. If the assumed depth is correct, the projected source pixels should correspond to the same physical scene point and should therefore have similar visual appearance.

Classical MVS methods evaluate this consistency using hand-designed matching costs. For example, they may compare pixel intensities, image patches, or local descriptors across views. A range of candidate depth values is tested for each reference pixel, producing a matching-cost volume. Each element of this volume describes how well the observations agree for one pixel and one depth hypothesis.

Estimating depth independently from the lowest matching cost is often unreliable. Image regions may be textureless, repetitive, occluded, reflective, or affected by illumination changes. Classical methods therefore combine the matching cost with spatial regularization. This encourages neighbouring pixels to have compatible depths while still allowing discontinuities at object boundaries. The final depth map is obtained by optimizing a combination of multi-view photometric consistency and spatial smoothness.

Learning-based MVS follows the same geometric principle but replaces several hand-designed components with neural networks. Instead of comparing raw image intensities, a convolutional feature extractor computes learned feature maps for each input view. These features can be trained to be more robust to local appearance changes, weak texture, noise, and illumination variation than direct pixel comparisons.

For a selected reference view, the source-view features are warped into the reference camera for a set of candidate depth hypotheses. Consider a reference pixel $\mathbf{x}$ and a candidate depth d . The pixel is first back-projected into three-dimensional space using the reference-camera intrinsics:

$$\mathbf{X}_r(d) = d\mathbf{K}_r^{-1}\tilde{\mathbf{x}}, \quad (2.6)$$

where $\tilde{\mathbf{x}}$ is the homogeneous image coordinate and $\mathbf{K}_r$ is the intrinsic calibration matrix of the reference camera. The point is then transformed into the coordinate system of a source camera:

$$\mathbf{X}_s(d) = \mathbf{R}_{sr}\mathbf{X}_r(d) + \mathbf{t}_{sr}, \quad (2.7)$$

where $\mathbf{R}_{sr}$ and $\mathbf{t}_{sr}$ describe the relative pose between the reference and source cameras. Finally, the point is projected into the source image:

$$\tilde{\mathbf{x}}_s(d) \sim \mathbf{K}_s\mathbf{X}_s(d). \quad (2.8)$$

Since the projected coordinates are generally not integer pixel locations, the source feature map is sampled using differentiable interpolation. Repeating this operation for all reference pixels and depth hypotheses produces a warped source feature volume aligned with the reference view.

The warped features from multiple source views must then be combined into a single cost representation. A common method for this is variance-based aggregation [11]. Let $\mathbf{F}_i^w(\mathbf{x}, d)$ denote the feature vector of view i aligned with reference pixel $\mathbf{x}$ at depth hypothesis d , where the reference feature volume is included as one of the N views. The aggregated cost $\mathbf{C}(\mathbf{x}, d)$ is defined by

$$\bar{\mathbf{F}}(\mathbf{x}, d) = \frac{1}{N} \sum_{i=1}^N \mathbf{F}_i^w(\mathbf{x}, d), \quad \mathbf{C}(\mathbf{x}, d) = \frac{1}{N} \sum_{i=1}^N \left(\mathbf{F}_i^w(\mathbf{x}, d) - \bar{\mathbf{F}}(\mathbf{x}, d) \right)^2 \quad (2.9)$$

where the square is applied element-wise to the feature channels. A low variance indicates that the features are similar and therefore supports the depth hypothesis. A high variance indicates disagreement between views.

The resulting tensor is referred to as a cost volume. It has two spatial dimensions, one depth dimension, and a feature dimension. It explicitly represents the matching evidence for every tested depth at every reference pixel. Because the volume is constructed using the known camera geometry, the network does not need to learn the complete relationship between camera motion and image correspondence from data.

The raw cost volume is still ambiguous and noisy. Learning-based MVS methods therefore use a three-dimensional convolutional network to regularize it. Three-dimensional convolutions jointly process the image dimensions and the depth dimension, allowing the network to combine local spatial context with neighbouring depth hypotheses. This helps suppress isolated matching errors, resolve ambiguous regions, and preserve coherent surfaces.

The regularized cost volume is converted into a probability distribution over the candidate depths, commonly using a softmax operation along the depth dimension. The depth for a reference pixel can then be selected as the most probable hypothesis or computed using the expectation

$$\hat{d}(\mathbf{x}) = \sum_{j=1}^D P(d_j | \mathbf{x}) d_j, \quad (2.10)$$

where D is the number of depth hypotheses and $P(d_j | \mathbf{x})$ is the predicted probability of depth d_j .

In summary, learning-based MVS combines explicit projective geometry with learned feature extraction and cost-volume regularization. The geometric warping determines which observations from different views should correspond under each depth hypothesis, while the neural components learn how to compare these observations and resolve ambiguous matches. This separation allows the method to use known camera calibration directly while remaining robust to challenging image regions such as weakly textured surfaces, repeated patterns, and partial occlusions.

2.5 Volumetric Fusion

A depth map provides a two-dimensional representation in which each valid pixel stores the distance from the camera to the observed surface. Using the camera intrinsics, every depth value can be back-projected into three-dimensional space. The corresponding camera pose then transforms the resulting points into a common world coordinate system.

A simple reconstruction approach would therefore convert each depth map into a point cloud and combine the point clouds from all viewpoints. However, this usually produces a noisy and redundant representation. The same surface is observed multiple times, and small depth errors cause the corresponding points to appear at slightly different positions. Furthermore, a point cloud contains samples of the surface but does not directly describe which regions of space are free, occupied, or located inside an object.

Volumetric fusion addresses these issues by representing the reconstruction on a regular three-dimensional grid. The reconstruction volume is divided into small cubic cells called voxels. Instead of storing only observed surface points, each voxel stores information about its position relative to the measured surfaces. Multiple depth observations can then be integrated into the same volume.

A widely used volumetric representation is the signed distance function (SDF). For a point $\mathbf{X}$ in three-dimensional space, the SDF stores the distance to the closest surface:

$$\phi(\mathbf{X}) = s(\mathbf{X}) \min_{\mathbf{Y} \in \mathcal{S}} \|\mathbf{X} - \mathbf{Y}\|_2, \quad (2.11)$$

where $\mathcal{S}$ denotes the set of all points on the scene surface and $s(\mathbf{X})$ determines the sign. A common convention assigns positive values to points in front of the surface, towards the observing camera, and negative values to points behind it. Points located directly on the surface have a signed distance of zero. The surface can therefore be recovered as the zero level set

$$\mathcal{S} = \{\mathbf{X} \mid \phi(\mathbf{X}) = 0\}. \quad (2.12)$$

In practice, the complete signed distance to the nearest surface is not estimated. Instead, a truncated signed distance function (TSDF) is used [1]. Distances are only stored within a limited region around the measured surface. Let μ denote the truncation distance. The truncated value is defined as

$$\phi_{\text{TSDF}}(\mathbf{X}) = \text{clip} \left(\frac{\phi(\mathbf{X})}{\mu}, -1, 1 \right), \quad (2.13)$$

where clip restricts the value to the interval $[-1, 1]$. Consequently, voxels close to the surface contain detailed distance information, while voxels farther away are assigned the constant value -1 or 1 . Truncation reduces the influence of measurements far from the surface and limits the region that must be updated for every depth image.

To integrate a depth map, the centre $\mathbf{X}_w$ of each relevant voxel is first transformed from world coordinates into the coordinate system of the current camera:

$$\mathbf{X}_c = \mathbf{R}_{cw} \mathbf{X}_w + \mathbf{t}_{cw}. \quad (2.14)$$

The voxel is then projected into the depth image using the camera intrinsics. For $\mathbf{X}_c = (X_c, Y_c, Z_c)^\top$, its image location is

$$\tilde{\mathbf{x}} \sim \mathbf{K} \mathbf{X}_c. \quad (2.15)$$

If the projected location lies inside the image and has a valid measured depth $d(\mathbf{x})$, the position of the voxel is compared with the observed surface along the camera ray. The projective signed distance is commonly computed as

$$\delta(\mathbf{X}_w) = d(\mathbf{x}) - Z_c. \quad (2.16)$$

The normalized truncated distance contributed by the current depth map is therefore

$$\hat{\phi}(\mathbf{X}_w) = \text{clip} \left(\frac{\delta(\mathbf{X}_w)}{\mu}, -1, 1 \right). \quad (2.17)$$

Each voxel stores both a TSDF value and an accumulated confidence weight. When a new measurement is available, it is combined with the previous estimate using a weighted average:

$$\Phi_{k+1}(\mathbf{X}) = \frac{W_k(\mathbf{X})\Phi_k(\mathbf{X}) + w_{k+1}(\mathbf{X})\hat{\phi}_{k+1}(\mathbf{X})}{W_k(\mathbf{X}) + w_{k+1}(\mathbf{X})}, \quad (2.18)$$

$$W_{k+1}(\mathbf{X}) = W_k(\mathbf{X}) + w_{k+1}(\mathbf{X})$$

where Φ_k is the current fused TSDF value, W_k is its accumulated weight, and $\hat{\phi}_{k+1}$ is the contribution from the new depth map. The measurement weight w_{k+1} may be constant or may depend on factors such as viewing angle, depth uncertainty, sensor confidence, or distance from the camera.

This repeated averaging is the main mechanism through which volumetric fusion reduces random depth noise. If several measurements observe the same surface from different viewpoints, their zero crossings are combined in the common volume. Random errors tend to average out, while consistently observed surface locations receive larger accumulated weights.

The voxel size determines the spatial resolution of the reconstruction. Smaller voxels can represent finer geometric details but increase memory usage and computation. The truncation distance also affects the result. A large truncation region produces smoother surfaces and allows measurements with larger errors to influence each other. A small truncation region preserves sharper detail but requires the input depth maps and camera poses to be more accurate.

After all depth maps have been integrated, a polygonal surface is extracted from the zero crossing of the fused TSDF. This is commonly performed using an isosurface extraction algorithm such as Marching Cubes [7]. The algorithm examines neighbouring voxel values and places surface vertices where the TSDF changes sign. The resulting vertices and triangles form a dense mesh representation of the reconstructed scene.

TSDF fusion depends strongly on the consistency of the input measurements. Random depth noise can often be reduced through averaging, but systematic errors remain problematic. If depth is consistently overestimated or underestimated, the reconstructed surface will be displaced. Inaccurate camera poses cause observations from different viewpoints to be integrated

at different world positions, which may create duplicated, thick, or blurred surfaces. Occlusions and missing depth values can additionally produce incomplete regions.

Chapter 3

Related Work

This chapter reviews prior work on learning-based multi-view stereo and event-based depth estimation. It then relates these approaches to the method proposed in this thesis and motivates the main design choices.

3.1 Image-Based Multi-View Stereo

Learning-based MVS methods for images retain an explicit separation between geometry and learning. Camera calibration and projective transformations determine how features are compared across views, while neural networks learn suitable features and regularize the resulting cost volume. This combination allows the models to exploit known multi-view geometry while learning to handle image ambiguities that are difficult to describe using fixed matching functions.

MVSNet was one of the first architectures to combine differentiable geometric warping, learned feature matching, cost-volume construction, and three-dimensional regularization in a fully end-to-end system [11]. However, constructing a cost volume at high spatial resolution over many depth hypotheses requires substantial memory. The cost grows approximately linearly with the number of depth samples and image pixels, while the three-dimensional regularization network introduces additional memory requirements.

CasMVSNet addresses this limitation using a coarse-to-fine cascade [4]. The first stage operates at a low spatial resolution and searches over the complete depth range. It produces an initial, coarse depth estimate. Subsequent stages operate at higher spatial resolutions but search only within a smaller interval around the previous prediction. This reduces the number of depth hypotheses required at high resolution.

CasMVSNet also uses a feature pyramid to provide image features at several spatial scales. Coarse features contain broader contextual information and are used for the initial global depth search. Finer features preserve local structure and object boundaries and are used to refine the estimate. The cascade therefore improves both memory efficiency and depth resolution.

3.2 Event-Based Depth Estimation

Event-based depth-estimation methods include learning-based monocular approaches, fixed-baseline stereo methods, and motion-based multi-view methods. These categories are not mutually exclusive, as recent approaches combine explicit multi-view geometry with learned depth estimation.

E2Depth demonstrated that dense metric depth can be predicted directly from monocular event streams [5]. It converts events into voxel grids and processes them using a recurrent U-Net-like architecture. Recurrent components allow information to be accumulated across consecutive event windows. Importantly, the network produces dense depth predictions, including at pixels with little or no recent event activity, by using spatial context and temporal information accumulated through its recurrent architecture.

Event-Based Multi-View Stereo (EMVS) recovers depth from a moving event camera with a known trajectory [9]. Similar to conventional RGB-based MVS, it evaluates candidate depths by using calibrated observations acquired from multiple viewpoints. However, rather than comparing image intensities, patches, or learned features across views, EMVS back-projects individual events as rays into a disparity-space volume and measures their spatial density. Rays generated by the same scene structure concentrate at the correct depth, allowing depth to be recovered from local maxima of the ray-density volume. EMVS is entirely geometry-based and contains no learned components. Because events and their corresponding rays are concentrated primarily around brightness edges, the resulting depth maps are semi-dense rather than dense.

Learning-based stereo methods have also shown that correspondences can be recovered from event data using neural networks. These methods typically use two synchronized event cameras with a fixed stereo baseline [8].

DERD-Net is described by its authors as the first learning-based multi-view stereo method for event-based depth estimation [6]. Like EMVS, it first uses known camera poses to back-project events into disparity-space images that encode ray densities. It then applies a neural network with three-dimensional convolutions and recurrent components to local subregions of these volumes, learning to identify depth from ray-density patterns instead of selecting maxima using a purely geometric rule. This improves both depth accuracy and completeness compared with geometry-only ray-density processing. However,

inference is still performed for selected pixels with sufficient ray-density evidence, so the resulting depth estimates remain semi-dense rather than covering the complete image.

3.3 Relation to This Thesis

E2Depth shows that dense depth can be learned from event voxel grids. EMVS shows that known camera motion can provide strong multi-view depth constraints for a single event camera. DERD-Net demonstrates that event data and known poses can be combined in a learning-based MVS system. MVSNet and CasMVSNet provide an established framework for dense feature-based multi-view depth estimation.

This thesis combines these directions in a controlled robotic reconstruction setting. The main components adopted from previous work are:

E2Depth: The use of spatiotemporal voxel grids for dense depth prediction from event data.

MVSNet: Feature warping, variance-based view aggregation, cost-volume construction, and three-dimensional regularization.

CasMVSNet: Multi-scale feature extraction and coarse-to-fine depth refinement.

TSDF fusion: The integration of predicted depth maps into a dense volumetric reconstruction.

Unlike DERD-Net, which estimates depth only at selected pixels supported by sufficient ray-density evidence, this thesis aims to predict dense depth maps despite the spatial sparsity of event data. E2Depth provides evidence that dense prediction from sparse event streams is possible by processing spatiotemporal voxel grids with a learned model. Motivated by this result, the thesis combines the voxel-grid input representation used by E2Depth with established components of learning-based image MVS. It then evaluates whether this combination can produce dense and accurate depth maps that can be used for three-dimensional reconstruction.

Chapter 4

Dataset Collection

To train and evaluate the proposed depth-estimation methods, we recorded a dataset containing synchronized event data, depth measurements, and calibrated camera poses. These observations provide the inputs and ground truth required for depth prediction and the subsequent multi-view reconstruction.

4.1 Recording System

The recording system consists of an Intel RealSense D435 depth camera and an IDS UE-39B0XCP-E event camera equipped with an IMX636 sensor. Both cameras are rigidly mounted on the end effector of a Franka robotic arm.

The RealSense D435 contains an RGB camera and an active-stereo depth system. Depth maps are recorded at a resolution of 640×480 pixels and a frequency of 30 Hz. The event camera captures events at a resolution of 1280×720 pixels. The two cameras are hardware synchronized using a trigger system.

The mount used to attach the two cameras to the robot end effector was 3D-printed. It keeps the relative arrangement of the cameras fixed throughout calibration and recording. The camera mount and the complete environment are shown in figures 4.1 and 4.2.

4.2 Suppression of Infrared Interference

The RealSense D435 uses an infrared projector to add a high-frequency texture pattern to the scene for active-stereo depth estimation. Although the projected pattern is not visible to the human eye, it is detected by the event camera. During camera motion, changes in the observed pattern generate a large number of events unrelated to the visible object geometry.

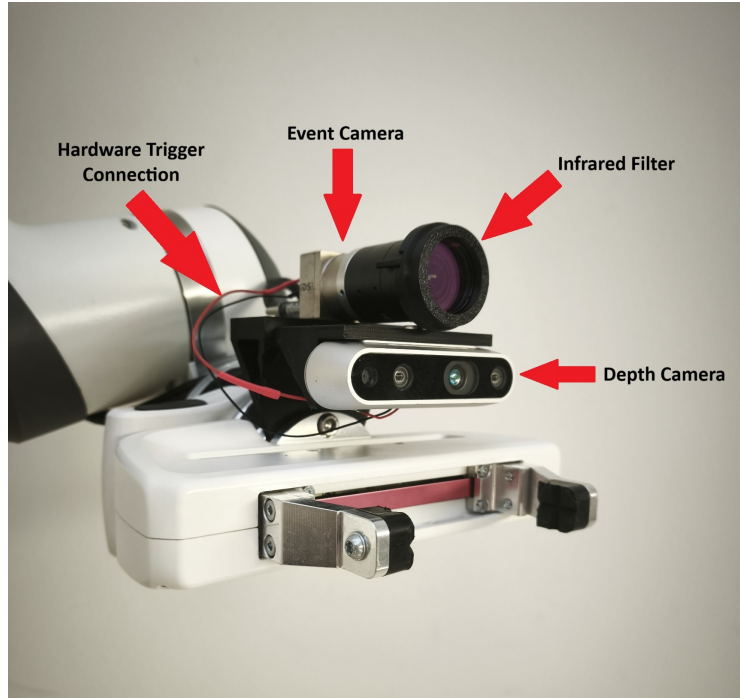


Figure 4.1: Close-up view of the robot arm end effector with the mounted cameras.

To suppress this interference, an SVBony 1.25-inch telescope filter is mounted in front of the event-camera lens. A circular clip was 3D-printed to attach the filter to the lens of the event camera. The filter suppresses the infrared pattern produced by the RealSense projector while preserving events generated by the object and the surrounding scene. After installing the filter, the projector pattern is no longer visible in the event stream.

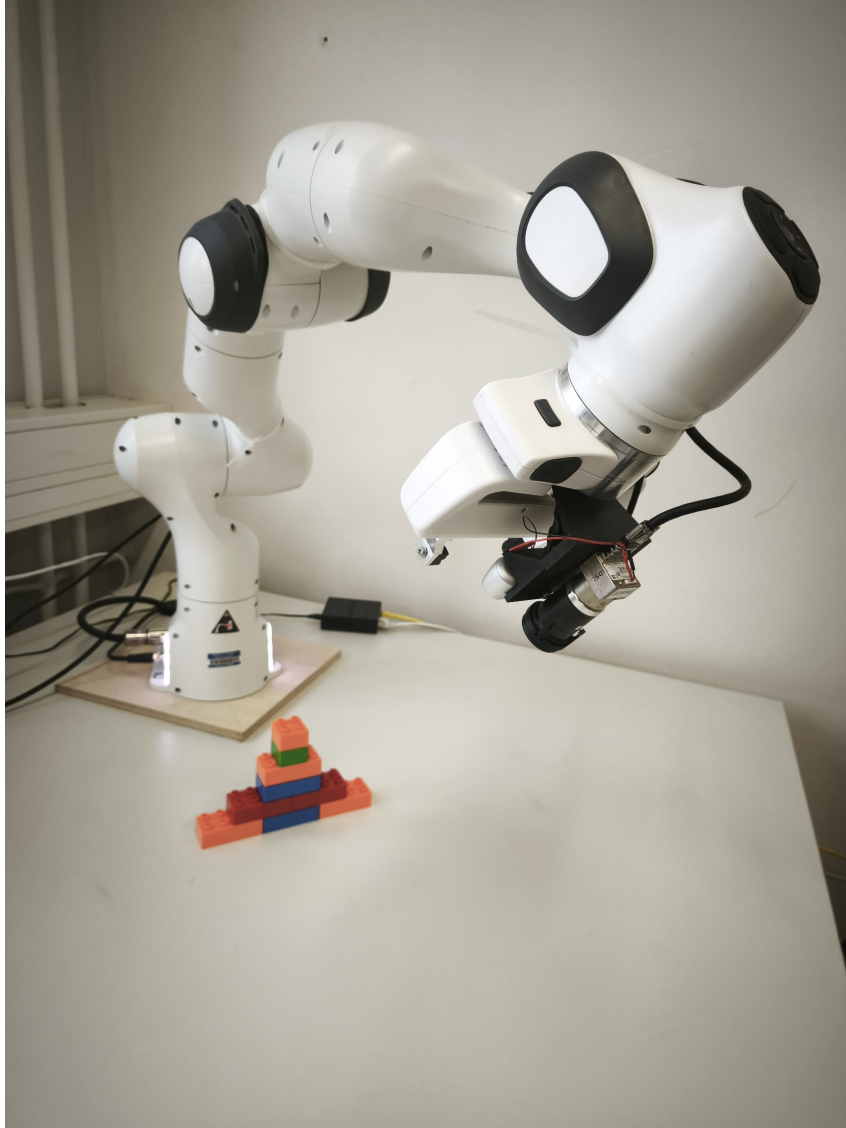


Figure 4.2: Complete experimental setup consisting of the Franka robotic arm, the mounted cameras, and the object recording area.

4.3 Camera Calibration

Calibration is required to project the RealSense depth measurements into the event-camera frame, thereby generating ground-truth depth maps, and to derive the event-camera poses from the robot configuration for use in the MVS pipeline.

A ChArUco [3] calibration board, is recorded from 60 different robot poses. The board is observed by both the RGB camera and the event camera. Recording the board from different positions and orientations provides the observations required to estimate the camera intrinsics and the rigid transformations between the cameras and the robot end effector.

Unlike a conventional camera, an event camera only generates events when the observed brightness changes. The calibration board would therefore normally have to move relative to the event camera to make its pattern visible. One possible approach would be to move the robotic arm and associate the recorded observations with the corresponding robot poses. However, the robot poses and camera observations would then have to be accurately associated while the robot is moving. Small synchronization errors would result in incorrect pose correspondences. Furthermore, continuous motion would cause motion blur in the RGB images, reducing the accuracy of ChArUco corner detection and consequently the accuracy of the calibration.

Therefore, a strobe light is used during calibration. The robot first moves to one of the calibration poses and remains stationary. The flashing illumination causes repeated brightness changes across the complete calibration board, which generates events even though the camera and board do not move. These events can be accumulated to produce an image-like representation of the board. Standard ChArUco corner detection can then be applied to both the RGB images and the event-camera observations.

The recorded calibration data is used to estimate the event-camera intrinsics and the rigid transformation between the RGB and event cameras. Hand-eye calibration is used to determine the transformation between the robot end effector and the RealSense RGB camera.

The RealSense depth-camera intrinsics and the rigid transformation between the depth and RGB cameras are obtained from the manufacturer calibration. The RGB intrinsics are instead estimated from the recorded ChArUco observations. The required transformations can therefore be obtained by combining

- the hand–eye transformation between the end effector and the RGB camera,
- the calibrated transformation between the RGB and event cameras,
- and the manufacturer-provided transformation between the RGB and depth cameras.

This transformation chain provides both the event-camera pose and the transformation required to project depth measurements into the event-camera coordinate system. The calibration setup is shown in figure 4.3



Figure 4.3: The calibration setup, consisting of the ChArUco board, the camera system, and the strobe light used to generate events without motion. The main coordinate frame is visible at the robot’s base.

4.4 Dataset Collection

A dedicated dataset was recorded to provide synchronized event observations, ground-truth depth maps, and camera poses. Static objects constructed from plastic building blocks were placed on a white table and observed from multiple viewpoints.

4.4.1 Recorded Objects and Conditions

The recorded objects are constructed from green, blue, orange, and red plastic building blocks and placed on a white textureless table. The sequences are recorded under similar indoor lighting conditions. During each sequence, the observed object remains static while the robotic arm moves the cameras around it. Two examples of the recorded objects are shown in Figure 4.4.

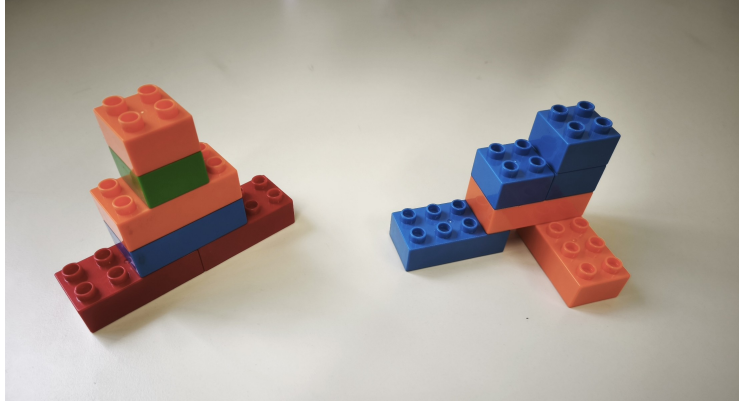


Figure 4.4: Two examples of recorded objects for the dataset.

In total, the dataset contains 48 object-specific recording sequences. Of the 48 recorded objects, 42 are assigned to the training set and 6 are reserved as a held-out validation set. The validation sequences were selected randomly. No separate test set is provided.

4.4.2 Viewpoint Sampling

A target point is defined slightly above the table at the expected centre of the object. End-effector positions are randomly sampled from a restricted hemisphere around this target. For every sampled position, the end-effector is oriented such that it points towards the target point. This ensures that the object is clearly visible to both cameras, as their optical axes are roughly aligned with the end effector’s z -axis. Furthermore, the camera’s x -axis is kept parallel to the supporting table plane. This keeps the object upright and close to the image center throughout the recording.

For each trajectory, 30 end-effector poses are sampled. The distance between the end effector and the target is restricted to a range of 0.22 m to 0.33 m.

Several additional constraints restrict the admissible part of the hemisphere. Positions farther than 0.52 m from the robot base are removed because they cannot be reached reliably. Positions within 0.38 m of the robot base are also excluded because they often require difficult or infeasible arm configurations. Poses that risk a collision with the table are rejected, and the upper part of the hemisphere is excluded to avoid unnecessary viewpoints directly above the object.

These restrictions result in a set of reachable end-effector poses covering different viewing angles and distances from the target. The resulting pose space provides the diversity required for multi-view depth estimation and three-dimensional reconstruction.

Figure 4.5 visualizes the restricted sampling region together with the event-camera positions recorded for the dataset.

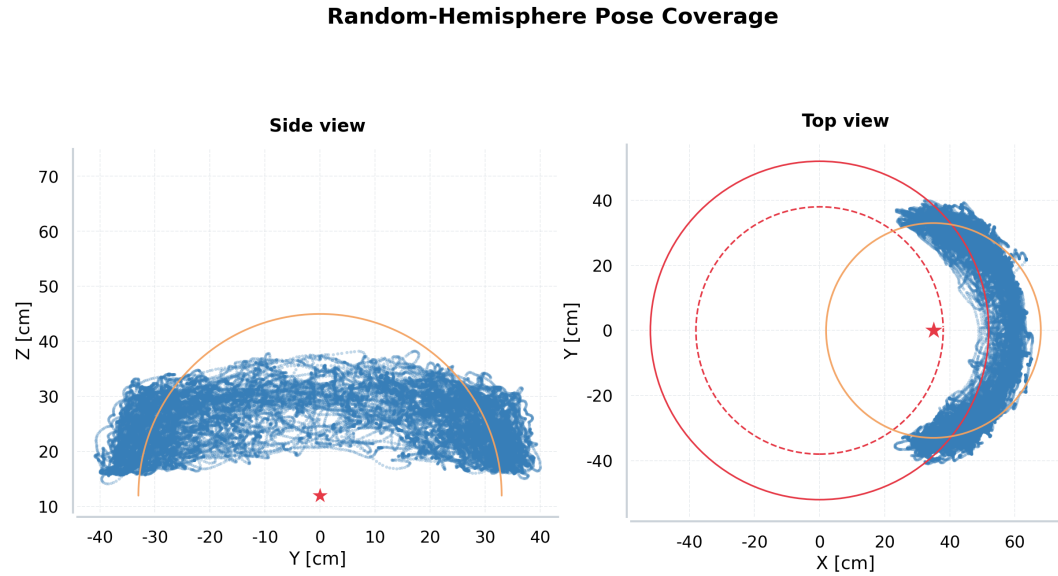


Figure 4.5: Visualization of the restricted hemisphere space in the base frame. The blue points represent all event-camera positions recorded for the dataset. The red lines indicate the restrictions made according to the robots base. The yellow line shows the outer hemisphere radius. The camera poses do not strictly align with the shown restrictions, as the cameras position is not equal to the end effectors position. The red star indicates the position of the chosen target point.

4.4.3 Trajectory Generation

The randomly sampled poses are reordered before execution to ensure a smooth movement path around the object. An initial path is constructed using a nearest-neighbour heuristic. Starting from the current pose, the closest unvisited target pose is repeatedly selected until all poses have been included.

The resulting path is subsequently improved using 2-opt local search. This procedure iteratively exchanges pairs of path edges whenever the exchange reduces the total path length. The resulting ordering provides a shorter and smoother traversal of the sampled viewpoints.

An example of one resulting camera trajectory is shown in Figure 4.6.

Data is recorded continuously while the robot follows the trajectory. Because the sampled poses and their ordering differ between sequences, the dataset contains variation in camera paths, viewing directions, and distances to the object.

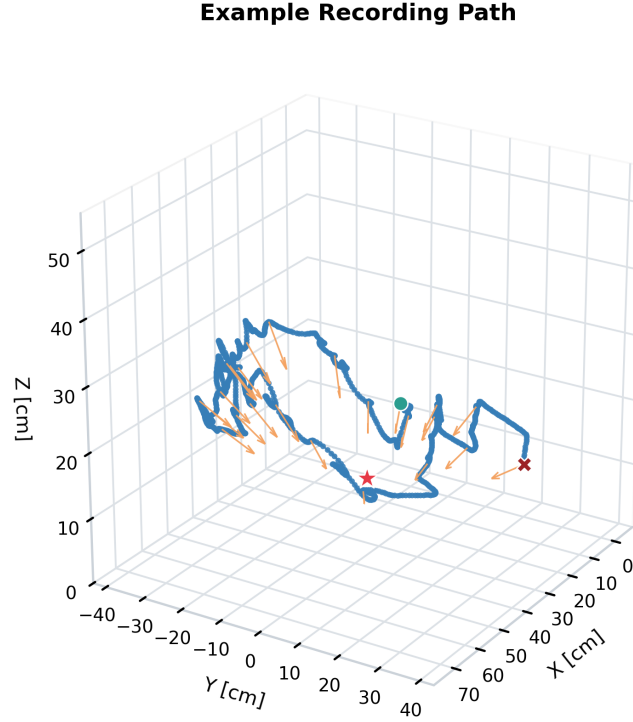


Figure 4.6: Visualization of an example event camera path for one recorded sequence. The blue line consists of the camera poses at each frame index. A yellow arrow shows the viewing axis of the event camera at each sampled position that defined the path. The red star indicates the target point position. The green circle shows the start position and the red x the end position of the sequence.

The robot controller also produces different camera velocities depending on the distance between consecutive target poses. Most recorded translational camera velocities range from 0 to 0.2 m/s as shown in figure 4.7.

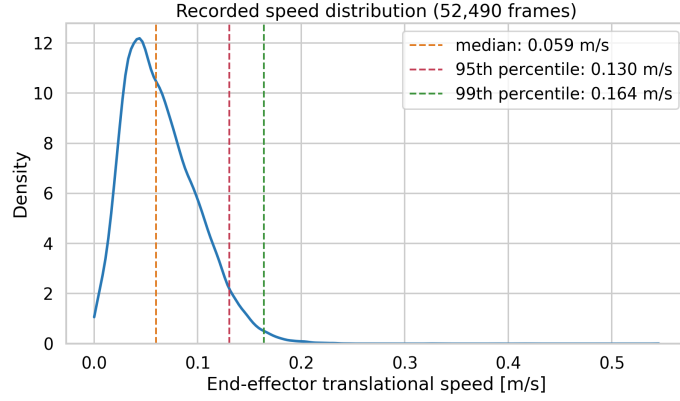


Figure 4.7: Distribution of the recorded end-effector velocities. The orange, red and green dashed lines visualize the mean, 95th percentile and 99th percentile of the velocity distribution

The resulting sequence lengths range from approximately 798 to 1590 frames. At the depth-camera rate of 30 Hz, this corresponds to recording durations of approximately 26 to 53 seconds. The duration is not fixed because it depends on the randomly sampled poses. Different pose configurations result in trajectories of different lengths and therefore require varying amounts of time for the robotic arm to complete.

This variation is particularly relevant for event data because the number and temporal distribution of generated events depend on the relative motion between the camera and the scene.

4.5 Data Preprocessing

The recorded data is precomputed into a format suitable for supervised depth prediction. The main preprocessing steps are the projection of depth into the event-camera frame, event voxelization, and spatial resizing and cropping.

4.5.1 Projection into the Event-Camera Frame

The RealSense depth maps are initially expressed in the depth-camera coordinate system. To align them with the event observations, depth pixels are projected into three-dimensional space using the depth-camera intrinsics.

The resulting points are transformed into the event-camera coordinate system using the calibrated transformation between the depth and event camera

and then projected into the event-camera image plane using the calibrated event-camera intrinsics.

The resulting depth map represents the scene from the same viewpoint as the event camera and can therefore be used as pixel-aligned ground truth for supervised training and evaluation. The later model operates within a depth range of $d_{\min} = 0.05$ m to $d_{\max} = 0.70$ m. This restricted range is appropriate for the object-centric recording setup because the aim is to predict the depth of the object and its immediate surrounding area, rather than distant parts of the scene or the background. Valid projected depth values are normalized to the interval $[0, 1]$ according to

$$\hat{d} = \text{clip} \left(\frac{d - d_{\min}}{d_{\max} - d_{\min}}, 0, 1 \right). \quad (4.1)$$

Invalid projected pixels remain excluded through the corresponding validity mask. Figure 4.8 shows an example frame of recorded and processed data.

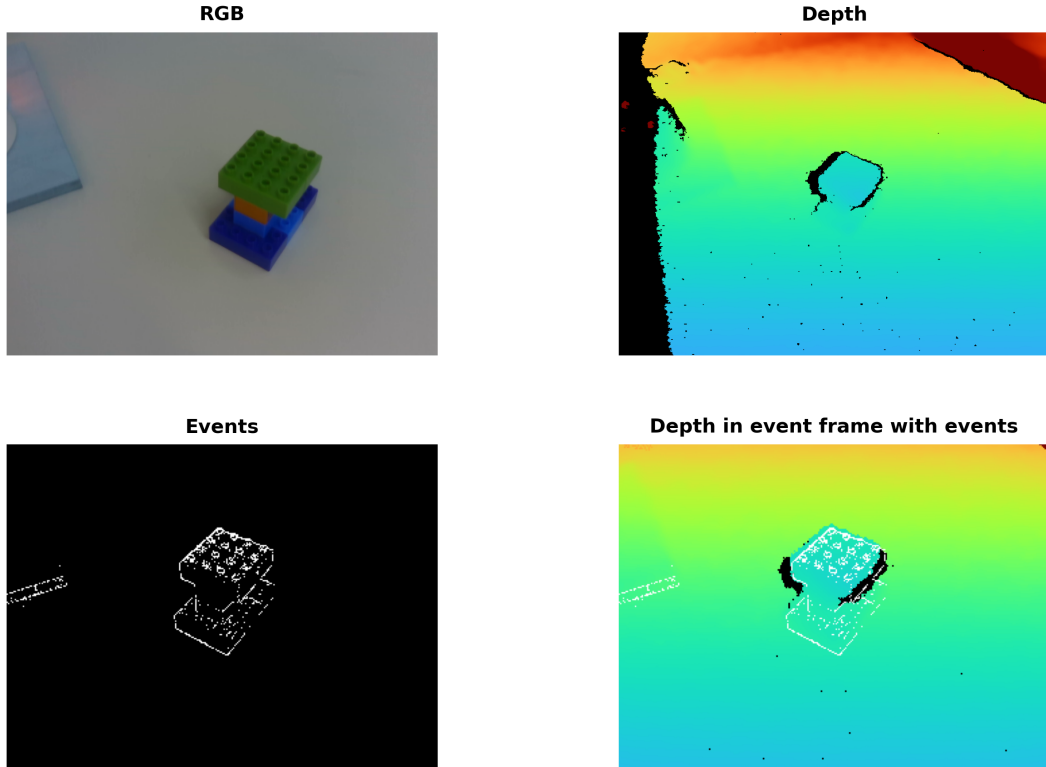


Figure 4.8: The different data modalities for one frame of the recorded data. Events are accumulated over the duration of one frame. Invalid depth is shown in black. The bottom-right image shows events and depth together in one frame after preprocessing was applied.

4.5.2 Event Voxelization

One event voxel grid is generated for each depth frame. The depth camera operates at 30 Hz, resulting in an approximate frame period of

$$\Delta t_{\text{frame}} = \frac{1}{30} \text{ s} \approx 33.33 \text{ ms.} \quad (4.2)$$

The depth and event streams are synchronized using hardware triggers. For each depth frame, let t_d^{trig} denote the timestamp of its corresponding trigger in the synchronized time base. This timestamp directly defines the centre of the event window, so no nearest-timestamp or nearest-interval selection is required. A voxelization window of approximately one depth-frame period is placed symmetrically around the trigger timestamp:

$$\left[t_d^{\text{trig}} - \frac{\Delta t_{\text{frame}}}{2}, t_d^{\text{trig}} + \frac{\Delta t_{\text{frame}}}{2} \right]. \quad (4.3)$$

The events inside this interval are represented using $B = 5$ temporal bins, following the voxel-grid construction described in Section 2.2. Specifically, each event timestamp is normalized to the temporal-bin coordinate according to Equation (2.3), and its polarity is distributed linearly between the two neighbouring temporal bins according to Equation (2.4). After accumulation, the active, nonzero entries of each voxel grid are standardized by subtracting their mean and dividing by their standard deviation. Inactive entries remain zero. This normalization reduces variation in the overall event magnitude between frames while retaining the spatial and coarse temporal structure of the event representation. Consequently, the voxel grid encodes event location, signed event contributions, and coarse temporal ordering while providing a fixed-size tensor of shape $5 \times H \times W$ for every depth frame.

4.5.3 Spatial Resizing and Cropping

The projected depth maps and event voxel grids are first centre-cropped from the native event-camera resolution of 1280×720 pixels to 960×720 pixels. The cropped representations are then resized to the network input resolution of 320×240 pixels.

The same resize and crop operations are applied to the event and projected depth data to preserve their pixelwise alignment. Reducing the spatial resolution lowers the storage requirements and the computational cost of training. The centred crop is suitable for the recording setup because both cameras are roughly directed towards the object, which is therefore expected to remain close to the centre of the image.

After preprocessing, each sample contains an event voxel grid, an aligned ground-truth depth map, an event-camera pose, and the adjusted camera intrinsics required by the depth-estimation pipelines.

4.5.4 Table-Plane Prior

The textureless white table produces few events away from object boundaries. Consequently, the event representation alone provides little information about the depth of the supporting surface. Moreover, the apparent depth of the table changes with the camera pose. A table-plane prior is therefore computed to provide the network with an explicit geometric reference for the static support surface.

The table is represented as a horizontal plane at the known height z_{table} in the robot-base coordinate system. For each frame, the calibrated event-camera pose and camera intrinsics are used to cast one ray through every image pixel. Let $\mathbf{o}$ denote the camera centre and $\mathbf{d}(u, v)$ the ray direction for pixel (u, v) , both expressed in the robot-base coordinate system. The intersection depth with the table plane is given by

$$\lambda(u, v) = \frac{z_{\text{table}} - o_z}{d_z(u, v)}. \quad (4.4)$$

This produces a dense image containing the camera-space depth of the prior plane at every pixel. Rays that do not intersect the plane in front of the camera are assigned the maximum model depth. The table-plane prior is clipped and normalized like the ground truth depth. It is computed using the camera intrinsics obtained after applying the centred crop and subsequent resizing used for the event data.

The prior does not contain measured object depth. It is derived only from an approximation of the table height, camera calibration, and robot pose. It can therefore also be generated during inference. It provides an absolute geometric reference where the event input is sparse and helps the network distinguish the support surface from objects extending above it. In particular, the network can learn depth as a deviation from the expected table geometry instead of inferring the supporting surface independently from sparse events in every frame.

The prior is not a perfect depiction of the table. In fact, because of imperfect calibration, it is not possible to give the exact table plane as prior for every camera position. The network can later learn to use the prior as a guideline and actively counter this imperfection.

Chapter 5

Methodology

This chapter describes the proposed learning-based event MVS method. Each observation consists of an event voxel grid, a table-plane prior, and a calibrated camera pose. For every reference observation, suitable source views are selected using the recorded camera poses. The reference observation and the selected source observations are then provided to the proposed learning-based event MVS pipeline, which predicts a dense reference-view depth map and a per-pixel confidence map. The chapter also presents the U-Net baselines used for comparison and the subsequent fusion of predicted depth maps into a three-dimensional reconstruction.

5.1 Multi-View Depth-Estimation Pipeline

The proposed model adapts the geometric cost-volume formulation of MVSNet [11] and the coarse-to-fine estimation strategy of CasMVSNet [4] to event-camera data. Instead of RGB images, the network processes event voxel grids together with an approximate supporting-plane prior. Given one reference observation and a set of temporally neighbouring source observations, it predicts a dense depth map in the coordinate system of the reference camera.

The pipeline comprises five main components: multi-view input construction, shared feature extraction, geometry-based feature warping, coarse-to-fine cost-volume processing, and full-resolution depth refinement. A separate head estimates the confidence of the final depth prediction. An overview is shown in Figure 5.1.

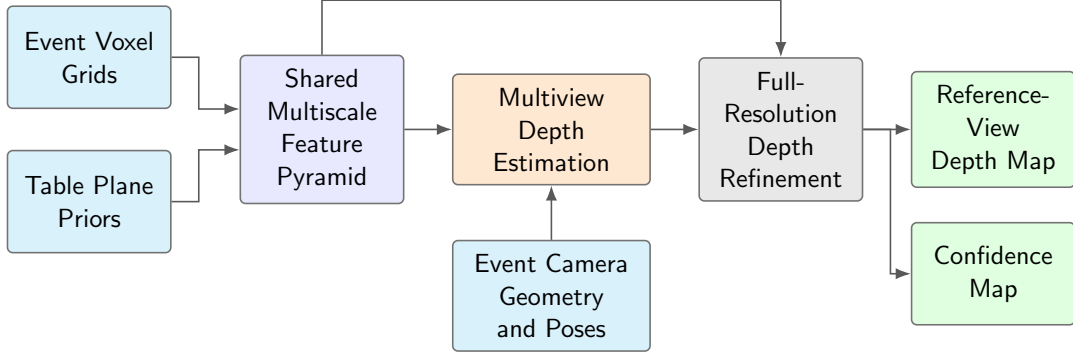


Figure 5.1: Overview of the proposed depth estimation pipeline. Event voxel grids and table plane priors are processed by a shared multiscale feature pyramid. Calibrated camera geometry supports multiview depth estimation, followed by full-resolution refinement and confidence prediction. Only the main data flow is shown.

5.1.1 Inputs and View Selection

As described in Section 2.4, conventional MVS estimates the depth of a reference image by comparing it with images of the same static scene captured from other viewpoints. The proposed method uses the same reference–source-view formulation, but replaces each RGB image with one event voxel grid. Thus, one voxel grid corresponds to one image, or one view, in a conventional RGB-based MVS pipeline.

For view i , the five-bin event voxel grid $V_i \in \mathbb{R}^{5 \times H \times W}$ is concatenated with the single-channel table-plane prior Π_i :

$$I_i = [V_i; \Pi_i] \in \mathbb{R}^{6 \times H \times W}. \quad (5.1)$$

The prior contains the expected depth at which each camera ray intersects the approximate table plane. It is generated for the pose of that particular view.

One event observation is designated as the reference view for which depth is predicted. Eight additional observations of the same static scene are used as source views. Four source views are selected before and four after the reference observation. Selection is based on camera displacement instead of a fixed temporal interval. Let $\tau_{\text{view}} = 0.05\text{m}$ denote the minimum translation required between consecutively selected views. Starting at the reference view, the recording is scanned independently backward and forward in time. In each direction, the first observation whose camera centre is at least τ_{view} from the current anchor is selected and becomes the new anchor. The scan then continues until up to four source views have been found in that direction. Distance is measured as the Euclidean distance between camera centres in the

common world coordinate system, so this selection criterion depends on camera translation rather than elapsed time or camera rotation. If fewer than four qualifying observations exist on one temporal side, only the available source views are used. Internally, missing source slots are masked in the geometric cost-volume model.

Consequently, one prediction uses up to nine views in total, consisting of one reference voxel grid and up to eight source voxel grids, each paired with its own table-plane prior.

The dataset provides the rigid transformation $\mathbf{T}_{i \leftarrow w} \in \text{SE}(3)$ from the world coordinate system to the camera coordinate system of each selected view. These transformations are used to compute the relative poses required for feature warping.

5.1.2 Shared Multi-Scale Feature Extraction

All views are processed independently by the same shared feature extractor,

$$\left(F_i^c, F_i^m, F_i^f\right) = E(I_i), \quad (5.2)$$

where c , m , and f denote coarse, middle, and fine feature resolutions.

The feature extractor follows a residual feature-pyramid design. A bottom-up pathway extracts increasingly contextual features at reduced spatial resolutions, while a top-down pathway combines these features with higher-resolution representations through lateral connections. The coarse representation contains the strongest semantic context, whereas the middle and fine representations retain progressively more spatial detail.

The extractor uses two-dimensional convolutions, with the five temporal event bins and the table-plane prior treated as input channels. The stem and the three bottom-up feature levels use the resolutions and feature dimensions listed in Table 5.1, with stride-two downsampling between successive levels.

Table 5.1: Configuration of the shared multiview feature pyramid.

Level	Resolution	Features
Stem	$H \times W$	16
Fine	$H/2 \times W/2$	32
Middle	$H/4 \times W/4$	64
Coarse	$H/8 \times W/8$	128

In the top-down pathway, coarse and middle feature maps are bilinearly upsampled by a factor of two and a 1×1 convolution halves their number of channels so that it matches the next finer bottom-up level. This convolution

operation can also be seen as a standard learned linear transformation of the features. After that, the resulting feature map is added element-wise to the corresponding one generated in the bottom-up path. Output convolutions refine the fused representations while preserving their dimensions. The resulting feature pyramid is shown in Figure 5.2.

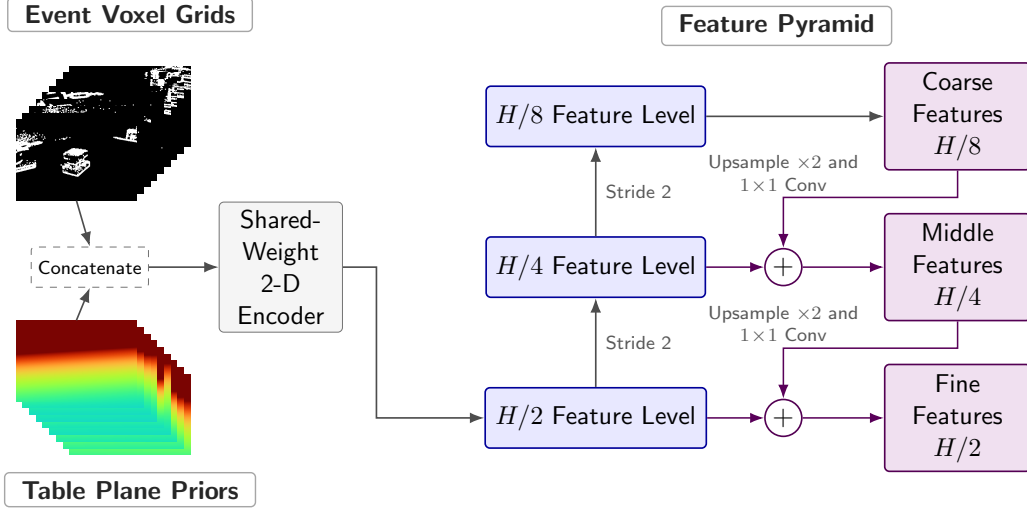


Figure 5.2: Detailed overview of the shared multiview feature pyramid. Each event representation is concatenated with the table-plane prior belonging to the same camera pose. Shared encoder weights produce a bottom-up feature hierarchy for every selected view. A top-down pathway combines semantic information from low-resolution features with higher-resolution lateral features, yielding one feature map for each stage of the depth cascade.

5.1.3 Feature Warping and Cost-Volume Construction

The general MVS principle behind geometric feature matching and cost-volume construction is introduced in Section 2.4. This section describes how that principle is implemented in the proposed model.

To compare the reference and source views geometrically, the model evaluates a set of candidate depths for every reference-view pixel. For each candidate, the source-view features are aligned with the reference view. The camera intrinsic matrix is scaled to the resolution of the feature maps used for this operation.

For each reference pixel $\mathbf{x}$ and candidate depth d , backprojection, transformation into the source-camera coordinate system, and projection into the source view are performed as defined in Equations (2.6)–(2.8). In the proposed model, the intrinsic matrices in these equations are scaled to the resolution of

the current feature maps. The required relative transformation from the reference camera to source view i is obtained from the recorded poses as

$$\mathbf{T}_{i \leftarrow r} = \mathbf{T}_{i \leftarrow w} \mathbf{T}_{r \leftarrow w}^{-1}, \quad (5.3)$$

whose rotation and translation provide $\mathbf{R}_{sr}$ and $\mathbf{t}_{sr}$ in Equation (2.7). As described in Section 2.4, the source feature map is then sampled bilinearly at the projected position.

A validity mask is generated for every projected point. A projection is valid only when the transformed point lies in front of the source camera and inside the source feature map. Invalid projections are excluded from the subsequent multi-view aggregation.

After warping, the aligned source features and the reference feature are aggregated separately for every candidate depth using their channel-wise variance according to Equation (2.4). A low variance indicates agreement between the views under the corresponding depth hypothesis, whereas a high variance indicates inconsistent alignment, occlusion, or ambiguous event features. The cost volume is formed by collecting these variance features over all pixels and candidate depths. The whole process can be seen in Figure 5.3

The aggregation accounts only for geometrically valid source projections. In addition to the variance features, the cost volume contains a learned small 16-channel projection of the reference feature, repeated for every depth hypothesis, and the fraction of source views with a valid projection at that pixel and depth. These additional channels are used by the regularization network described in Section 5.1.4.

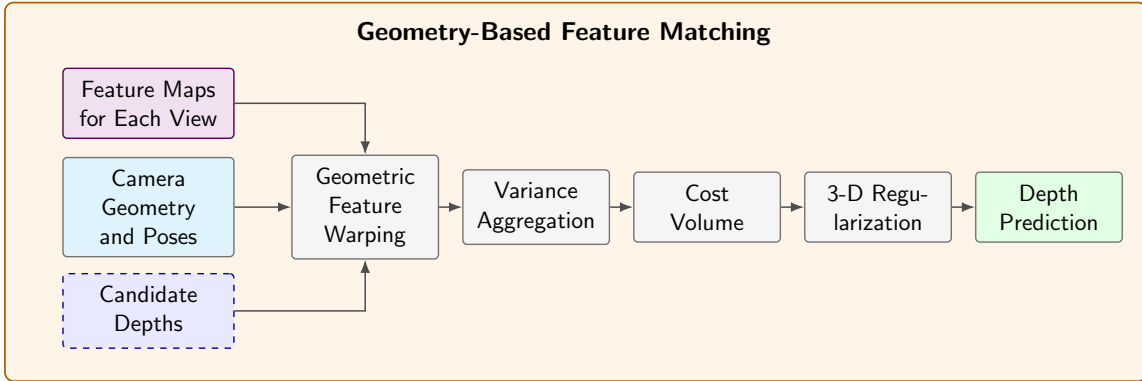


Figure 5.3: Geometry-based feature matching and cost-volume processing. Source features are warped into the reference view for the candidate depths and aggregated into a cost volume. A 3-D regularization network processes the volume and produces a depth map at the spatial resolution of the input feature map.

5.1.4 Cost Regularization and Depth Regression

The cost volume is processed by a three-dimensional U-Net-like regularizer with additive skip connections. Its convolutions operate jointly over the two spatial dimensions and the depth-hypothesis dimension, allowing the network to combine local image structure with consistency across neighbouring depth hypotheses. At each of the coarse, middle, and fine stages, the regularizer first maps its stage-specific cost-volume input to a base width of 64 cost channels. Deeper encoder levels increase the channel width as the volume resolution is reduced.

In addition to the variance features, the regularizer receives the compressed reference representation and the proportion of geometrically valid source projections stored in the cost volume. The reference representation preserves information about the reference view that is not captured by inter-view variance alone. The validity proportion indicates how many source views support each hypothesis, allowing the network to distinguish well-supported matches from hypotheses based on only a small number of valid projections.

The regularizer follows an encoder-decoder structure and is represented by the 3-D regularization block in figure 5.3. The encoder gradually reduces the spatial and depth-hypothesis resolutions so that evidence can be combined over a larger neighbourhood. The decoder restores the original volume resolution through trilinear interpolation. At each corresponding resolution, the encoder features are added element-wise to the decoder features rather than concatenated as in the original U-Net architecture. This preserves high-resolution encoder information without increasing the decoder channel width. It therefore limits the memory consumption and computational cost of the subsequent convolutions, which is particularly important because their feature maps extend across three dimensions, including both spatial dimensions and the depth-hypothesis dimension.

The regularizer assigns one scalar cost to every candidate depth. These costs are converted into a depth estimate using the probability-based depth regression described in Equation 2.10. Applying this regression independently at every reference-view pixel produces a complete depth map at the spatial resolution of the processed cost volume.

5.1.5 Coarse-to-Fine Depth Estimation

The feature-matching pipeline shown in Figure 5.3 produces a depth map from one set of feature maps and candidate depths. The model applies this pipeline three times at progressively higher spatial resolutions. These successive applications form the coarse, middle, and fine stages of the depth cascade and use the corresponding feature maps from the shared feature pyramid.

At each resolution, source features are warped for the current depth candidates and aggregated into a cost volume. After regularization, the candidate costs are converted into a probability distribution and regressed to a depth map as described in Section 2.4. The resulting map is bilinearly upsampled and provides the per-pixel centre for the candidate depths of the next stage. The connection between the three applications is illustrated in Figure 5.4.

This hierarchy allows the low-resolution coarse stage to search the complete depth range at manageable computational cost. Once an approximate depth has been found, the higher-resolution stages can concentrate fewer hypotheses in local intervals around that estimate. They therefore recover progressively finer spatial detail without constructing a large full-range cost volume at high resolution.

Coarse stage. At resolution $H/8 \times W/8$, the coarse stage evaluates 32 candidates over the complete depth range $[0.05, 0.70]$ m. The same candidate set is used at every pixel. The candidates are uniformly spaced in inverse depth. Inverse-depth spacing places candidates more densely at short distances, where a given metric-depth change causes a larger image displacement. The resulting probability distribution produces the coarse depth map $\hat{d}_c$.

Middle stage. The coarse map $\hat{d}_c$ is bilinearly upsampled to $H/4 \times W/4$. For each pixel p , 16 candidates are placed uniformly in a fixed metric window of ± 0.12 m around its upsampled coarse depth. Candidates outside the valid depth range are clipped to its boundaries. Unlike the global coarse candidates, these hypotheses therefore differ between pixels. Regularizing the resulting middle-stage cost volume and regressing its probabilities gives the refined map $\hat{d}_m$.

Fine stage. The middle map $\hat{d}_m$ is bilinearly upsampled to $H/2 \times W/2$. A small convolutional head receives this upsampled map and the fine-scale reference features and predicts a value that determines the local candidate spacing. Eight candidates are distributed uniformly over a learned interval around each pixel’s middle-stage estimate. The interval has a half-width between 0.02 and 0.08 m and is clipped to the valid depth range. The fine stage can therefore use a narrow interval where only a small correction is required and a wider interval where the previous stage is less certain. The probability-weighted regression of these candidates produces the half-resolution fine depth map $\hat{d}_f$, which is subsequently upsampled and processed by the full-resolution refiner, explained in the next Section 5.1.6.

Table 5.2: Configuration of the three-stage depth cascade.

Stage	Resolution	Hypotheses	Search interval
Coarse	$H/8 \times W/8$	32	$[0.05, 0.70]$ m
Middle	$H/4 \times W/4$	16	± 0.12 m
Fine	$H/2 \times W/2$	8	learned, ± 0.02 – 0.08 m

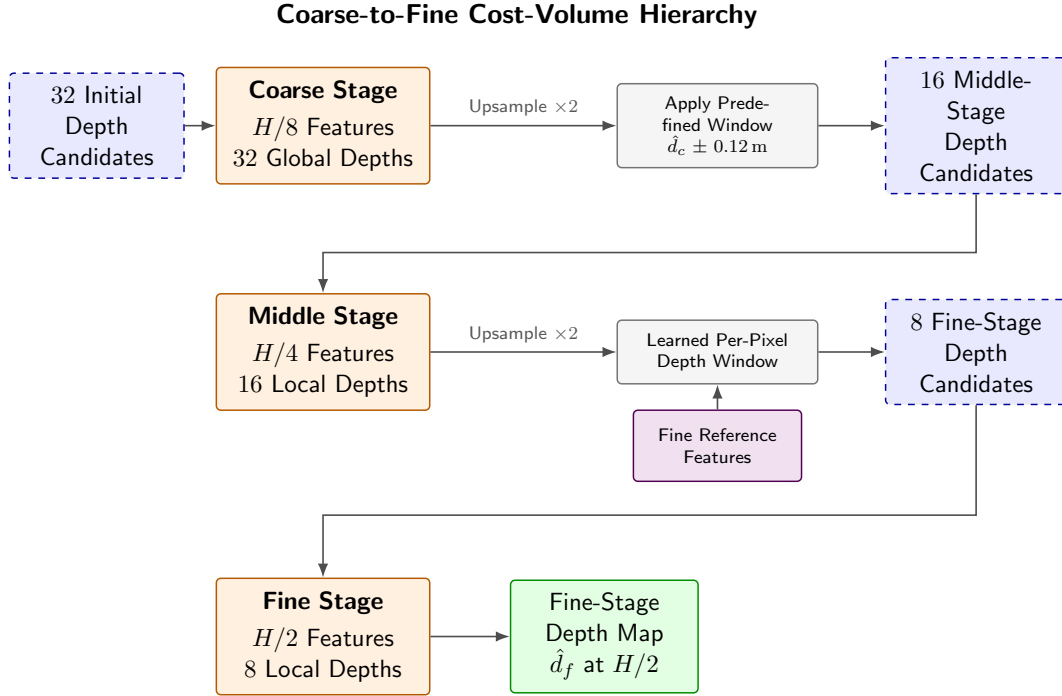


Figure 5.4: Detailed coarse-to-fine cost-volume hierarchy. Each orange stage applies the complete feature-warping, cost-volume, regularization, and depth-regression process shown in Figure 5.3. The coarse stage evaluates 32 initial candidates over the complete depth range and outputs a depth map at $H/8$ resolution. This depth map is bilinearly upsampled by a factor of two, and a fixed window around the upsampled values defines 16 candidates for the middle stage. The middle stage similarly outputs an $H/4$ depth map, which is bilinearly upsampled before a learned per-pixel window generates eight candidates for the fine stage. The hierarchy ends with the depth map $\hat{d}_f$ at $H/2$ resolution, generated by the fine stage.

5.1.6 Full-Resolution Refinement

The fine-stage prediction is produced at half the input resolution. It is bilinearly upsampled and passed to a two-dimensional residual refinement network together with the original six-channel reference-view input I_r defined in Equation 5.1 and the fine reference-view feature map F_r^f produced by the shared feature pyramid. The fine feature map is also bilinearly upsampled from half resolution. The refiner predicts a bounded residual correction that is added to the upsampled multi-view estimate. The result is then clipped to the valid depth range.

This stage does not perform additional geometric warping. Its purpose is to recover fine object boundaries and local detail from the reference observation while keeping the prediction anchored to the depth estimated from multiple views. The complete path from the fine-stage prediction to the final depth and confidence maps is summarized in Figure 5.5.

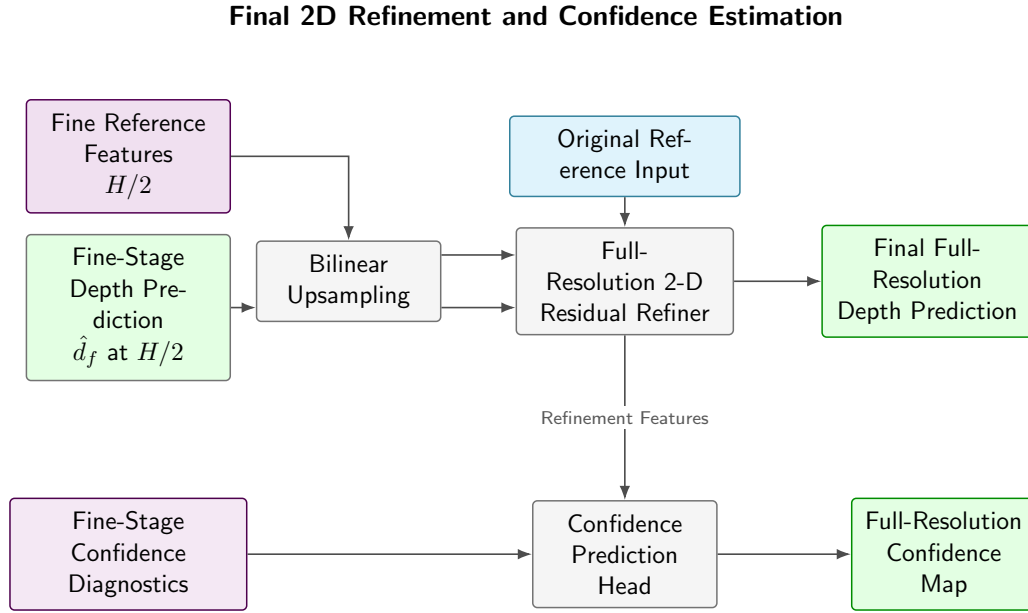


Figure 5.5: Full-resolution refinement and confidence estimation. The fine-stage depth prediction and fine reference features are bilinearly upsampled and combined with the original full-resolution reference event and plane-prior input by a 2-D residual refiner. Fine-stage confidence diagnostics are concatenated with the refinement features to predict the confidence map.

5.1.7 Confidence Estimation

A separate prediction head estimates a confidence value $c(\mathbf{x}) \in [0, 1]$ for each output pixel. Its purpose is to identify depth estimates that can be integrated reliably during TSDF reconstruction. The head combines features from the full-resolution refiner with compact diagnostics derived from the fine-stage probability volume and multiview matching process.

The probability volume is the same distribution used for fine-stage depth regression. It is summarized by its per-pixel standard deviation, entropy, and maximum probability. These values describe the concentration and ambiguity of the predicted depth distribution. The matching diagnostics additionally include the proportion of valid source projections and the disagreement between the middle- and fine-stage depth predictions. The diagnostic maps are brought to the output resolution and combined with the refiner features by a small prediction head with a sigmoid output.

The confidence target is a binary label indicating whether the final metric depth estimate at a pixel is accurate to within 1 cm. It is defined as

$$t(p) = \mathbb{I}\left[|\hat{D}(p) - D(p)| < \tau_{\text{conf}}\right], \quad \tau_{\text{conf}} = 1 \text{ cm}. \quad (5.4)$$

Here, p denotes a pixel, $\hat{D}(p)$ is the predicted depth, and $D(p)$ is the corresponding ground-truth depth. The absolute difference $|\hat{D}(p) - D(p)|$ is therefore the metric depth error at that pixel. The indicator function $\mathbb{I}[\cdot]$ returns one when the condition inside the brackets is true and zero otherwise. Consequently, $t(p) = 1$ marks a prediction whose absolute error is smaller than the confidence threshold τ_{conf} , while $t(p) = 0$ marks a prediction with an error that is at least equal to it.

Because accurate and inaccurate pixels can occur with different frequencies, the confidence prediction is supervised with class-balanced binary cross-entropy. Positive targets are weighted using the ratio of negative to positive valid pixels, preventing the more frequent class from dominating the confidence loss $\mathcal{L}_{\text{conf}}$.

The resulting confidence map is used during 3D reconstruction. Further details are provided in Section 5.2

5.1.8 Training Objective

The main training objective is to minimize the difference between the final depth prediction and the ground-truth depth map at the input image resolution. However, the projected ground-truth depth map is not defined at every pixel. Missing values can arise, for example, from occlusion, absent depth-camera returns, or regions that are not covered during projection into the event-camera image. These pixels must not contribute to the training loss. A validity mask $M(p) \in \{0, 1\}$ therefore indicates whether ground-truth depth is available at pixel p , and $N = \sum_p M(p)$ denotes the number of valid pixels. Before the loss is evaluated, predicted and ground-truth depths are normalized to the network depth range $[0, 1]$ using $d_{\min}$ and $d_{\max}$.

The primary depth term is a masked Charbonnier loss. With the normalized residual $r(p) = \hat{d}_n(p) - d_n(p)$, it is defined as

$$\mathcal{L}_{\text{depth}} = \frac{1}{N} \sum_p M(p) \sqrt{r(p)^2 + \varepsilon^2}, \quad \varepsilon > 0. \quad (5.5)$$

It behaves approximately like an absolute-error penalty, but remains differentiable even for small residuals.

Additionally, the multi-scale gradient loss compares changes in the depth residual in the horizontal and vertical image directions. This comparison is performed at the full output resolution and at three successively downsampled resolutions. At each scale, only neighbouring pixels with valid ground truth contribute. The absolute gradient differences are averaged across directions and scales. This encourages the prediction to reproduce local surface slopes and depth discontinuities rather than matching only the absolute depth values.

Finally, the surface-normal loss evaluates the orientation of the predicted surface. Predicted and ground-truth depths are first converted back to metres and backprojected into three-dimensional points using the camera intrinsics. Neighbouring points define horizontal and vertical surface directions, whose cross product provides a normal vector at each pixel. The loss averages the cosine distance between corresponding predicted and ground-truth normals over valid pixels. It therefore penalizes incorrect local surface orientation even when the absolute depth error is small.

The confidence term $\mathcal{L}_{\text{conf}}$ is the class-balanced binary cross-entropy between the predicted confidence and the target defined in Equation 5.4. The complete training objective used in the final configuration is

$$\mathcal{L} = \mathcal{L}_{\text{depth}} + \lambda_{\text{grad}} \mathcal{L}_{\text{grad}} + \lambda_{\text{normal}} \mathcal{L}_{\text{normal}} + \lambda_{\text{conf}} \mathcal{L}_{\text{conf}}. \quad (5.6)$$

All stages of the proposed architecture are supervised end-to-end through this final objective. The coarse, middle, and half-resolution fine-stage predictions are not supervised separately.

5.1.9 Data Augmentation and Training Regularization

Training-time augmentation is applied to the event channels while leaving the deterministic plane prior unchanged. The augmentations include removal of complete source observations, additive noise on active event voxels, shared event dropout across views, and rectangular masking of individual views. These transformations simulate missing events, sensor noise, and partial visibility, thereby discouraging the model from relying on individual events or source views.

Additional regularization is applied inside the network. Two-dimensional dropout is used on the feature-pyramid outputs and reference features, while three-dimensional dropout is applied at the bottlenecks of the cost-volume regularizers. Stochastic path dropping is further applied to feature-pyramid fusion paths. Together with AdamW weight decay, these operations reduce co-adaptation and overfitting to the comparatively small recorded dataset.

An exponential moving average (EMA) of the network parameters is maintained and is used for validation and checkpointing. During development, the unaveraged model exhibited inconsistent behaviour with pronounced loss spikes. EMA does not alter the gradient updates, but it smooths short-term parameter fluctuations and substantially stabilizes the model used for evaluation and final checkpoint selection.

5.2 Three-Dimensional Reconstruction

The depth networks predict one depth map in the event-camera image plane for each selected reference frame. To turn these independent predictions into a single three-dimensional reconstruction, the depth maps are first placed in the common robot base coordinate system and are then fused into a truncated signed distance function (TSDF) volume. For the reported reconstructions, 200 frames are selected at approximately uniform intervals over each sequence.

5.2.1 Backprojection and Pose Alignment

Let $D_i(u, v)$ be the predicted metric depth at pixel (u, v) in frame i and let $\mathbf{K}$ denote the event-camera intrinsic matrix after the resizing and cropping described in Section 4.5. A valid depth sample is backprojected into the coordinate system of camera i according to

$$\mathbf{X}_i(u, v) = D_i(u, v) \mathbf{K}^{-1} \begin{bmatrix} u & v & 1 \end{bmatrix}^T. \quad (5.7)$$

The calibrated robot pose provides the transformation $\mathbf{T}_{w \leftarrow i}$ from the event camera to the common robot-base coordinate system. Consequently, the cor-

responding world point is

$$\tilde{\mathbf{X}}_w(u, v) = \mathbf{T}_{w \leftarrow i} \tilde{\mathbf{X}}_i(u, v), \quad (5.8)$$

where the tilde denotes homogeneous coordinates. Applying the pose belonging to every frame aligns observations acquired from different viewpoints.

5.2.2 TSDF Integration

The aligned depth maps are integrated using the TSDF formulation and weighted update equations described in Section 2.5. The implementation uses a scalable voxel volume with a voxel side length of 2 mm and a truncation distance of 8 mm, corresponding to four voxel widths.

For multiview predictions, the estimated confidence map controls the observation weight in the fusion update. Because the employed legacy TSDF implementation does not accept a continuous per-pixel weight map, the continuous weight is approximated using eight nested confidence masks. Their thresholds are placed at the midpoints of eight equal subdivisions between a lower confidence bound of 0.2 and one. Each pixel is integrated once for every threshold its confidence exceeds, resulting in an effective discrete weight between zero and eight. Low-confidence pixels are omitted and neither update the surface nor carve free space, while high-confidence pixels contribute repeatedly.

After the standard zero-level-set extraction described in Section 2.5, vertex normals are computed. Ground-truth depth maps are fused separately with the same geometry and volume parameters to obtain a reference mesh for the reconstruction evaluation. For the final reconstructions, only the largest connected surface is retained. This removes small disconnected components that can appear as floating pieces above the object because of artifacts in the ground-truth depth data.

5.3 Evaluation

The proposed model is evaluated against a simpler U-Net baseline on the held-out validation set. No separate test split is used. In addition to standard depth-estimation metrics, object-focused metrics are used to prevent the comparatively easy table region from dominating the evaluation.

5.3.1 U-Net Comparison Baseline

Two U-Net configurations are included to evaluate the benefit of the proposed geometry-based multiview pipeline. The single-view configuration receives only the reference observation. The nine-view configuration receives

the reference and up to eight source observations. Its source observations are selected using the same pose-based procedure as for the proposed model. Although both configurations use the same network architecture, they are trained completely independently, with separate parameter initializations and optimization runs.

For each selected view i , the U-Net input contains

$$I_i = [V_i; \Pi_i; Q_i] \in \mathbb{R}^{12 \times H \times W}, \quad (5.9)$$

where V_i contains the five temporal event bins, Π_i is the table-plane prior, and Q_i contains six spatially constant pose channels encoding the camera position and optical-axis direction. For the multiview configuration, the inputs of all selected views are concatenated channel-wise before being passed to the network. The nine-view input retains a fixed layout comprising the reference, four past-source slots, and four future-source slots. If fewer than eight pose-selected source observations are available, the complete input block of each missing source slot is zero-padded; it is not replaced with an observation from the opposite temporal direction.

The baseline U-Net follows the main principles explained in Section 2.3 and its detailed implementation is shown in Figure 5.6. An initial convolution produces the first feature representation. Each encoder block applies a strided convolution for downsampling followed by a second convolution for local feature processing; batch normalization and ReLU activation follow both convolutions.

At the lowest-resolution stage, the latent representation is processed by two residual blocks. Each block contains two convolutions and adds its input to the processed features. Each decoder block bilinearly upsamples the current features, concatenates them with the corresponding encoder skip features, and processes the combined tensor with two convolutions, again followed by batch normalization and ReLU activation. The resolutions and feature dimensions are given in Figure 5.6.

A final 1×1 convolution and sigmoid activation produce a normalized depth prediction $\hat{d}_n \in [0, 1]$. The U-Net uses the same three depth-related terms as the multiview objective in Equation 5.6, but does not include a confidence prediction:

$$\mathcal{L}_{\text{U-Net}} = \mathcal{L}_{\text{charb}} + \lambda_{\text{grad}} \mathcal{L}_{\text{grad}} + \lambda_{\text{normal}} \mathcal{L}_{\text{normal}}. \quad (5.10)$$

The single-view baseline tests depth prediction from one observation, whereas the nine-view baseline tests whether the same encoder–decoder architecture can benefit from multiple pose-selected observations through direct channel-wise fusion. Unlike the proposed MVS model, neither variant constructs depth hypotheses, warps features, or forms cost volumes. Consequently, the nine-view U-Net must learn from the concatenated inputs how observations from different viewpoints and their associated camera poses should be combined.

Comparing the proposed model with both variants therefore separates the benefit of naively supplying additional observations from the benefit of explicitly incorporating multiview geometry.

U-Net Baseline Architecture

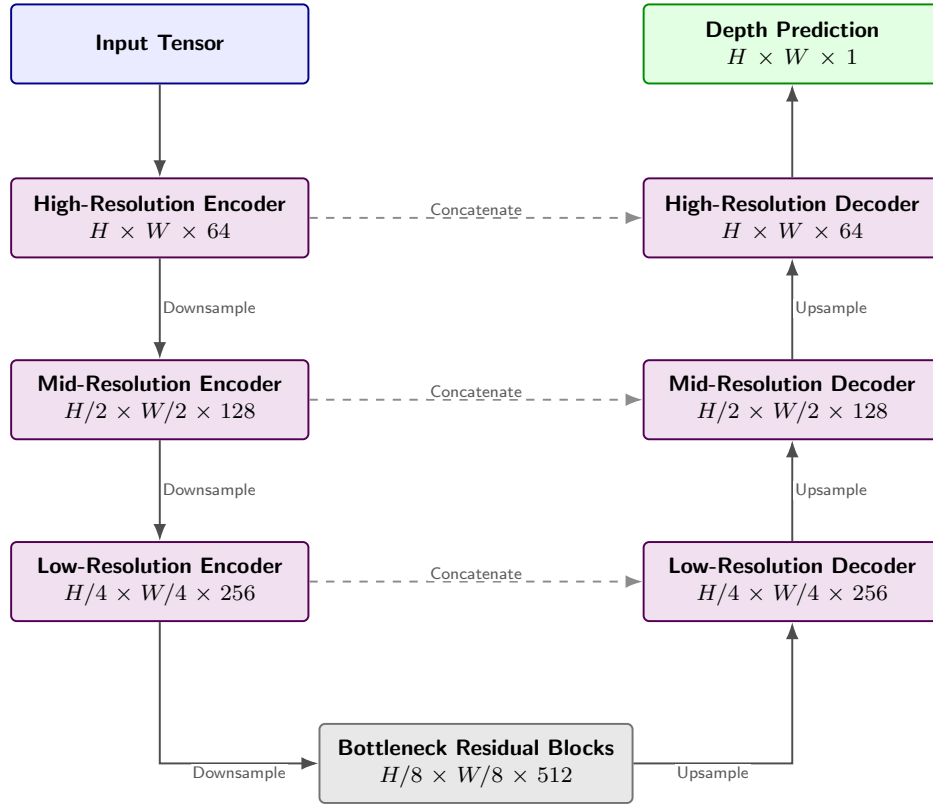


Figure 5.6: Architecture of the single-view and nine-view early-fusion U-Net baselines. The two configurations share the same encoder–decoder architecture and differ only in the number of input channels. The encoder progressively reduces the spatial resolution while increasing the feature width until the latent representation is reached. The decoder restores the input resolution through three upsampling stages and combines its features with encoder features at the corresponding resolutions through skip connections. A final prediction layer produces the dense depth output.

5.3.2 Object-Focused Error Metrics

The depth predictions can extend into large parts of the surrounding scene that are not relevant to the object-reconstruction task. We therefore define a 32 cm axis-aligned workspace cube in the robot-base coordinate system. The cube is centred horizontally around the known object position and extends from the table region to above the object. It contains the object and its nearby supporting surface while excluding the robot-arm base and other unrelated scene geometry.

An image-space mask is constructed from the ground-truth depth map. Each valid ground-truth pixel is backprojected using the event-camera intrinsics, transformed into the robot-base coordinate system using the recorded pose, and retained if the resulting point lies inside the cube. This construction makes the mask independent of the evaluated prediction. Errors between predicted and ground-truth depth can be reported inside this workspace mask. The workspace can be seen for an example view in figure 5.7.

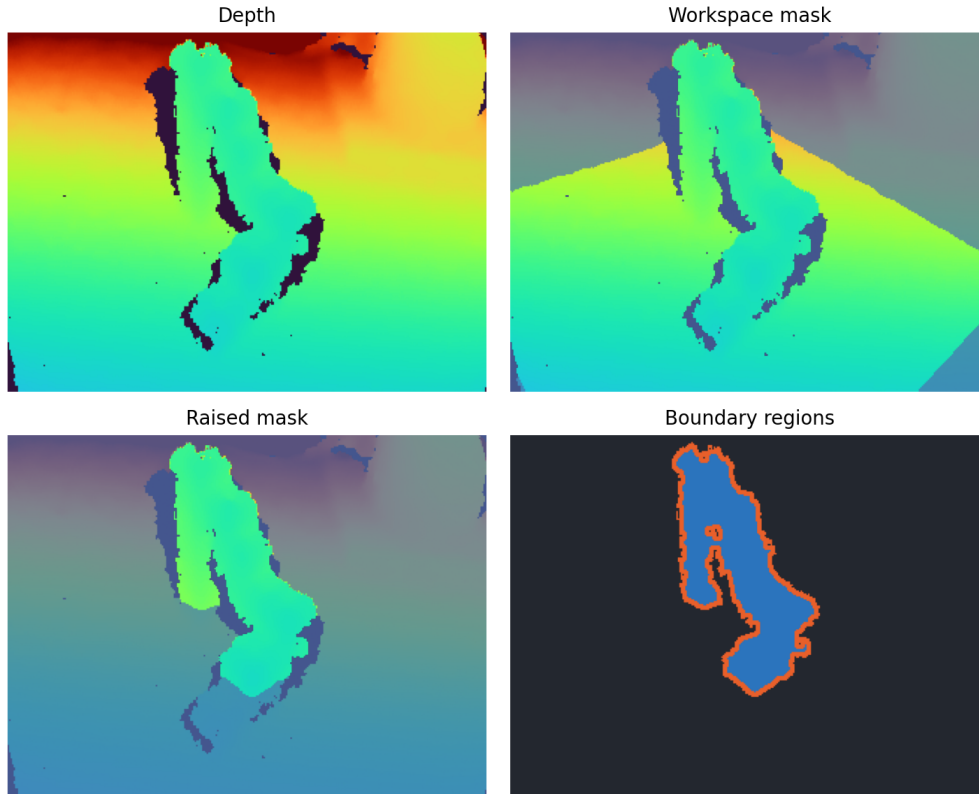


Figure 5.7: Object-focused evaluation masks. The unmasked depth map is shown together with the lower workspace mask, the raised object-focused mask, and the subdivision of the raised mask into boundary and non-boundary regions.

The same cube is also used to crop the predicted and ground-truth three-dimensional reconstructions, ensuring that their geometric comparison covers the same object-centred workspace.

A large part of this workspace can still be occupied by the supporting table, which is expected to be predicted comparatively well from the supplied table-plane prior. Errors on the object may therefore be underrepresented even within the workspace mask. To measure the object more directly, a second mask uses the same horizontal extent and upper limit, but raises the lower face to approximately 1.5 cm above the nominal table level. It removes the supporting plane and is used to report depth errors primarily on the object. While also excluding some pixels belonging to the object’s lower regions, it generally retains a large part of the object and therefore provides a useful metric that is substantially less influenced by the easily predicted table plane.

For the raised mask, boundary and non-boundary errors are reported separately. A retained pixel is initially classified as a boundary pixel when one of its four directly adjacent neighbours lies outside the raised ground-truth mask or has no valid ground-truth depth. This boundary is dilated by two pixels to cover a narrow region around the transition. All other valid pixels inside the raised mask form the non-boundary region. The two measurements distinguish errors near the object silhouette and missing-depth transitions from errors on the object’s interior surfaces. Figure 5.7 illustrates the two masks and this separation.

Chapter 6

Results

This chapter first describes the experimental setup and then reports the results in three main parts. First, the proposed geometry-based multiview model is evaluated as the primary MVS depth-estimation method, including its accuracy across evaluation regions and sequences, its qualitative predictions, and its confidence estimates. Second, the predicted depth maps are assessed in the downstream task of three-dimensional reconstruction. Finally, the multiview model is compared with the single-view U-Net and the nine-view early-fusion U-Net.

6.1 Experiments

The models were trained on 42 sequences and evaluated on 6 held-out sequences, which were never used for parameter updates. Ground-truth depth is represented in the event-camera frame, and the prediction range is defined as 0.05–0.70 m.

All models were trained for up to 50 epochs with a batch size of 14. They use the AdamW optimizer with an initial learning rate of 10^{-4} , weight decay 5×10^{-4} , and cosine decay to 10^{-6} without learning-rate warm-up. The training objective uses weights $\lambda_{\text{grad}} = 0.1$ and $\lambda_{\text{normal}} = 0.1$ for the gradient and the surface-normal.

The MVS model additionally introduces the confidence term 5.1.7 with weight $\lambda_{\text{conf}} = 0.1$ in addition to the primary depth loss, and the binary confidence target uses an absolute-error threshold of 1 cm. Dropout with probability 0.1 is applied to the feature-pyramid outputs, reference features, cost-volume regularizers, and stochastic path dropping.

An exponential moving average of the parameters with decay 0.9995 is used for validation and checkpointing for all three models. The reported checkpoint for each model is the EMA checkpoint with the lowest validation MAE.

Training was performed on a single NVIDIA L40S GPU. During final evaluation, all models use identical target frames, ground truth, camera calibration, and validity masks. Table 6.1 summarizes the resulting comparison.

Table 6.1: Models used in the final comparison. Parameter counts include all trainable parameters.

Model	Views	Input channels	Parameters
Single-view convolutional encoder-decoder	1	12	24.09 M
Nine-view convolutional encoder-decoder	9	108	24.24 M
Geometry-based multiview	9	9×6	27.10 M

6.2 MVS Depth-Prediction Results

Depth accuracy is reported in the three spatial domains introduced in Section 5.3.2. The *whole-frame* domain contains every pixel with valid ground-truth depth. The *workspace cube* contains the object its surrounding supporting table. Finally, the *raised cube* shifts the lower face of this volume above the table plane and therefore focuses the evaluation on the object.

The reported metrics are mean absolute error (MAE), root mean squared error (RMSE), and absolute relative error (AbsRel). The threshold accuracies δ_1 , δ_2 , and δ_3 report the percentage of predictions that agree with the ground truth within increasingly permissive multiplicative factors of 1.25, 1.5625, and approximately 1.95. Lower values are better for MAE, RMSE, and AbsRel, while higher threshold accuracies are better. The complete multiview results for each domain are reported in Table 6.2.

Table 6.2: Depth-estimation results of the proposed geometry-based multiview model across the three evaluation regions.

Region	MAE [cm]	RMSE [cm]	AbsRel	δ_1 [%]	δ_2 [%]	δ_3 [%]
Whole-frame	0.182	0.687	0.00431	99.91	99.98	100.00
Workspace cube	0.180	0.630	0.00498	99.89	99.98	100.00
Raised cube	0.522	1.547	0.01492	99.36	99.89	99.98

One example depth prediction and its corresponding absolute-error map can be seen in Figure 6.1.

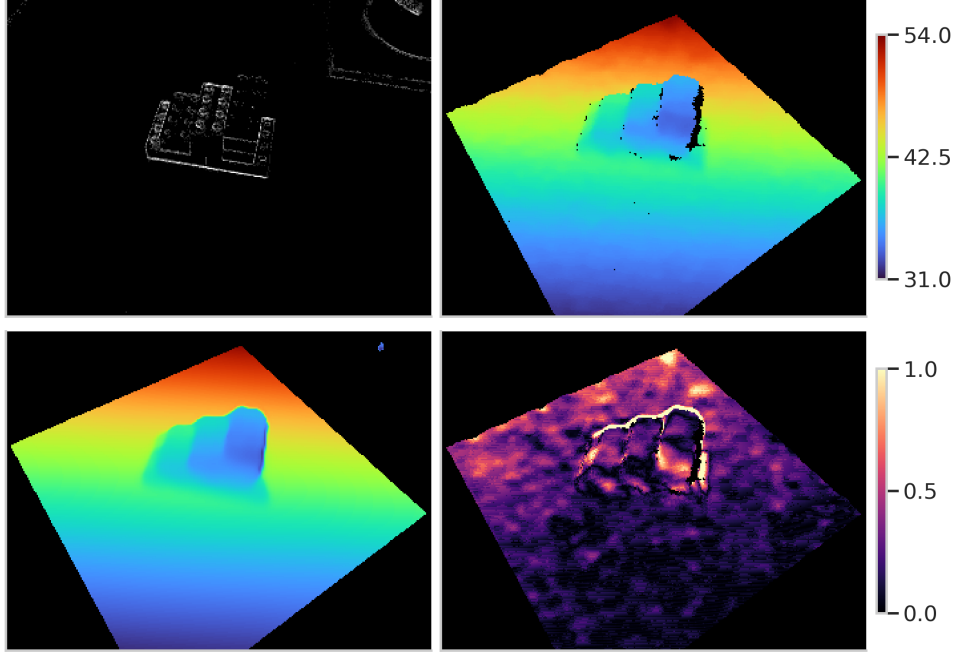


Figure 6.1: Events (top left), ground truth depth (top right), predicted depth (bottom left), and the respective absolute-error map (bottom right). Legend scale values are given in cm.

Removing the dominant table surface reveals a substantially more difficult object-focused prediction problem. The raised-cube MAE of 0.522 cm is approximately 2.9 times the whole-frame MAE of 0.182 cm and the workspace-cube MAE of 0.180 cm. The similar whole-frame and workspace-cube errors indicate that both aggregates are dominated by the comparatively easy supporting surface, whereas the raised cube exposes the larger errors on the object itself. Table 6.3 further separates the raised cube into the boundary and non-boundary regions defined in Section 5.3.2.

Table 6.3: Depth-estimation errors of the geometry-based multiview model in the boundary and non-boundary regions of the raised cube.

Region	MAE [cm]	RMSE [cm]
Boundary	0.940	2.542
Non-boundary	0.303	0.517

Boundary pixels have an MAE of 0.940 cm, compared with 0.303 cm away from boundaries, making the boundary error approximately 3.1 times larger. This gap shows that the remaining object-focused errors are concentrated particularly strongly around object edges. It is also the reason for the relatively high RMSE of 1.55 cm that was measured for the raised cube.

6.2.1 Performance Across Evaluation Sequences

Aggregate metrics can conceal substantial variation between objects and camera trajectories. Figure 6.2 therefore reports MAE separately for each held-out sequence and spatial domain, evaluated using the proposed multiview architecture. Some sequences are visibly more challenging than the others. This is predominantly caused by varying complexity between the objects. The resulting increase in error is hardly visible in the whole-frame and workspace-cube domains because both are dominated by the comparatively large and simple table surface. In the raised cube, which suppresses most of this supporting surface and concentrates the evaluation on the object, the higher error of Sequence 5 becomes pronounced. This contrast demonstrates why reporting multiple spatial domains is important. Whole-frame or workspace-level aggregates alone can conceal object-specific depth-estimation difficulties.

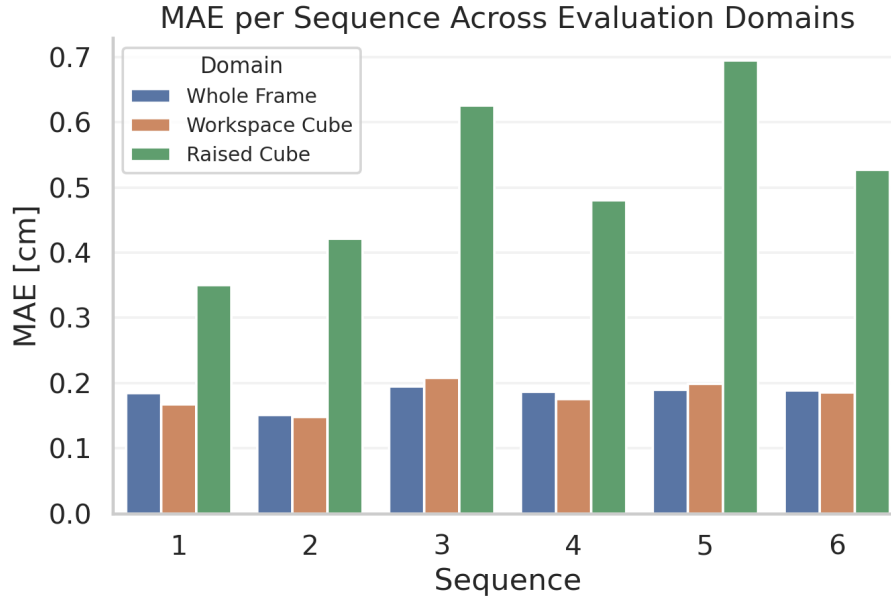


Figure 6.2: MAE per evaluation sequence for all three mask domains.

6.2.2 Confidence Prediction

The geometry-based multiview model additionally predicts a per-pixel confidence intended for confidence-weighted TSDF integration. Its usefulness must therefore be evaluated by its relationship to the actual depth error, rather than by visual appearance alone. A clear relationship is visible between confidence and prediction quality. Most pixels with high depth error receive low confidence, whereas pixels with high confidence generally have lower error. The confidence estimate therefore contains useful information about the reliability of the predicted depth.

Figure 6.3 supports the intended interpretation of confidence as an indicator of prediction reliability. If the predicted confidence actually improves TSDF fusion is discussed in section 6.3.1.

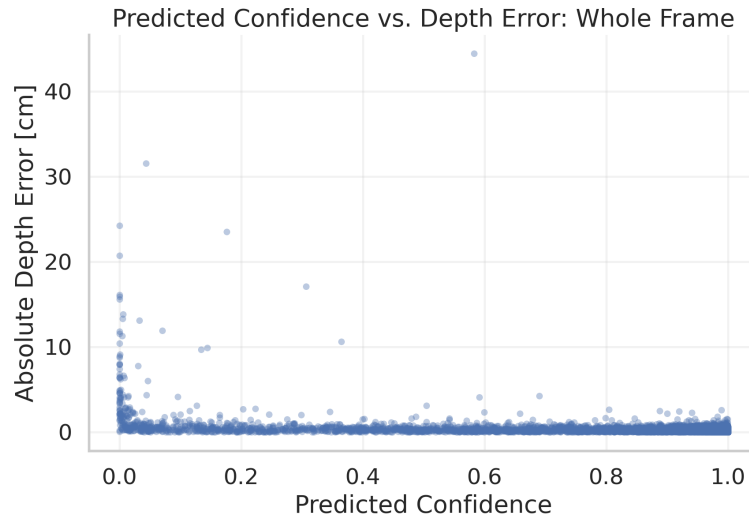


Figure 6.3: Absolute depth error over predicted confidence for twenty thousand randomly sampled pixels from the evaluation set. The whole frame domain was used, so no masking was applied. Most pixels with significant error are grouped in the low confidence section.

6.3 Three-Dimensional Reconstruction

Depth accuracy alone does not establish that predictions are consistent between viewpoints. The depth maps of the proposed multiview model are therefore fused using the TSDF reconstruction settings described in Section 5.2. The predicted meshes are compared with a reference mesh produced from the precomputed ground-truth depth maps. Figure 6.4 shows one such reference mesh next to the corresponding mesh reconstructed from predicted depth.



Figure 6.4: Example TSDF reconstructions generated from the ground truth depth maps (left) and depth map predictions from the MVS model (right).

The meshes are evaluated separately inside the workspace cube and the raised cube. For each evaluation region, 100,000 points are sampled uniformly from each of the predicted and reference surfaces. Accuracy is calculated by finding, for every point sampled from the predicted surface, the nearest point on the reference surface and averaging these distances. It therefore measures how closely the reconstructed geometry agrees with the reference and penalizes misplaced surfaces. Completeness is calculated in the opposite direction. For every point sampled from the reference surface, the distance to the nearest point on the predicted surface is averaged. It therefore measures how fully the reference geometry is represented and penalizes missing surfaces and holes. Lower values are better for both metrics. Their symmetric mean is reported as the Chamfer distance. F-scores additionally measure the balance between precision and recall at fixed distance tolerances, by computing their harmonic mean. The main results use the confidence-weighted integration described in Section 5.2.2.

The reported values in table 6.4 are means over the six reconstructed objects. The confidence-weighted reconstruction achieves a mean symmetric Chamfer distance of 0.108 cm inside the workspace cube. Its accuracy and completeness are closely balanced, indicating that the reconstructed surface is both near the reference surface and largely present in the prediction. This is also reflected by the 99.62 % F-score at a tolerance of 1 cm and by a normal consistency of 0.966.

Table 6.4: Average reconstruction metrics across the six evaluation sequences

Metric	Workspace cube	Raised cube
Accuracy [cm]	0.108	0.183
Completeness [cm]	0.108	0.177
Chamfer distance [cm]	0.108	0.180
Normal consistency	0.966	0.905
F@1 cm [%]	99.62	99.09

Table 6.5 reports evaluation metrics generated by rendering the reconstructed meshes from the evaluation views and comparing the rendered depth with the ground truth. We measure an MAE of 0.384 cm. The substantially higher RMSE of 1.162 cm is consistent with a small number of large disagreements near object boundaries. Depth errors at these discontinuities are often not consistent across viewpoints and are therefore largely averaged out during TSDF integration instead of forming a persistent displaced surface. When the resulting mesh is projected back into an evaluation view, however, even a small silhouette displacement can compare foreground depth with background depth at a few pixels. These large residuals receive disproportionate weight in the squared-error term and consequently raise RMSE much more strongly than MAE.

Table 6.5: Rendered-depth metrics for the reconstructed meshes.

Metric	Workspace cube	Raised cube
MAE [cm]	0.384	0.859
RMSE [cm]	1.162	2.743
AbsRel	0.0101	0.0205

6.3.1 Effect of Confidence-Weighted Fusion

Table 6.6 compares uniform TSDF integration with confidence-weighted integration for the geometry-based multiview model. Results are reported for the workspace cube and raised cube. To show the effect directly, the table gives the difference between the two modes error and their relative percentage change. Negative changes indicate an improvement for error metrics, whereas positive changes indicate an improvement for normal consistency and the F-score.

Table 6.6: Effect of confidence-weighted TSDF fusion. The absolute change Δ is the confidence-weighted result minus the uniform-fusion result, and the relative change is reported as a percentage of the result achieved without confidence usage.

Metric	Workspace Δ	Workspace change [%]	Raised Δ	Raised change [%]
Surface accuracy [cm]	-0.0059	-5.20	-0.0220	-10.77
Surface completeness [cm]	-0.0029	-2.63	-0.0102	-5.44
Chamfer distance [cm]	-0.0044	-3.93	-0.0161	-8.22
Normal consistency	+0.0047	+0.49	+0.0169	+1.91
F@1 cm [pp]	+0.0900	+0.09	+0.2998	+0.30

The relative changes show that confidence weighting has a meaningful effect on the extracted surface. Inside the workspace cube, surface accuracy improves by 5.2 %, completeness by 2.6 %, and Chamfer distance by 3.9 %. The improvements are larger in the raised cube, where the corresponding reductions are 10.8 %, 5.4 %, and 8.2 %. Normal consistency and the F-score also increase in both regions, although their relative changes are smaller because their uniform-fusion values are already close to one. These results indicate that the predicted confidence is useful for suppressing unreliable depth contributions and improving the extracted surface, especially in the object-focused region.

6.4 Comparison with the U-Net Baselines

The final part compares the proposed geometry-based multiview model with the single-view U-Net and the nine-view early-fusion U-Net. All three models are evaluated on the same six sequences using the same ground truth and spatial masks. This comparison separates the benefit of additional observations from the benefit of explicitly aligning those observations using camera geometry.

6.4.1 Depth Accuracy Across Evaluation Regions

Table 6.7 reports the main depth metrics for all three architectures in the whole-frame domain, workspace cube, and raised cube.

Table 6.7: Depth-estimation results for all architectures across the three evaluation regions.

Region	Model	MAE [cm]	RMSE [cm]	AbsRel	δ_1 [%]	δ_2 [%]	δ_3 [%]
Whole-frame	Single-view U-Net	0.228	0.869	0.00552	99.87	99.97	99.99
	Nine-view early-fusion U-Net	0.230	0.883	0.00558	99.86	99.97	99.99
	Geometry-based multiview	0.182	0.687	0.00431	99.91	99.98	100.00
Workspace cube	Single-view U-Net	0.262	0.904	0.00724	99.83	99.98	100.00
	Nine-view early-fusion U-Net	0.269	0.936	0.00741	99.82	99.97	100.00
	Geometry-based multiview	0.180	0.630	0.00498	99.89	99.98	100.00
Raised cube	Single-view U-Net	1.272	2.359	0.03633	99.07	99.81	99.96
	Nine-view early-fusion U-Net	1.340	2.447	0.03820	98.99	99.80	99.96
	Geometry-based multiview	0.522	1.547	0.01492	99.36	99.89	99.98

In the whole-frame domain, both U-Net baselines have only slightly larger errors than the geometry-based multiview model. The difference becomes clearer inside the workspace cube and substantial inside the raised cube. In the latter domain, the errors of both U-Nets are roughly two and a half times those of the geometry-based model. Figure 6.5 summarizes the MAE progression across the three regions.

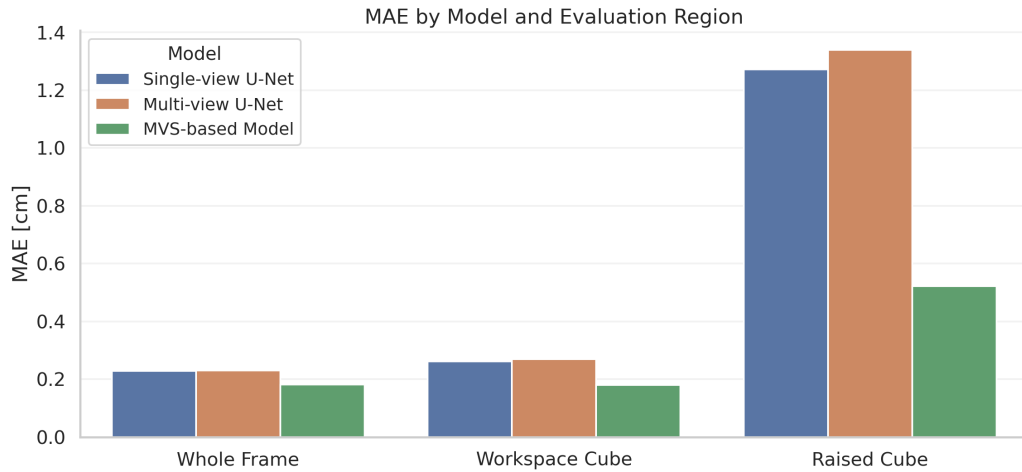


Figure 6.5: MAE for each evaluation region and each depth-estimation architecture.

Additional observations without geometric alignment. Providing nine observations to the early-fusion U-Net does not improve accuracy over the single-view model. Its error is instead slightly higher. Merely concatenating pose-selected views and pose channels therefore does not allow this architecture to exploit the additional observations effectively. A plausible interpretation is that the network must implicitly learn both correspondence and pose-conditioned alignment from channel-wise inputs. The extra channels then add variability without supplying the inductive bias needed to combine the observations.

Benefit of explicit multiview geometry. The geometry-based model receives the same number of observations as the nine-view early-fusion baseline but explicitly warps source features onto depth hypotheses before aggregating them. Its clear advantage, particularly inside the raised cube, indicates that the improvement arises from how the observations are aligned and compared rather than from their number alone.

6.4.2 Boundary and Non-Boundary Accuracy

Figure 6.6 compares the MAE in both the boundary and non-boundary region of the raised cube mask. For all three architectures, the error is higher near object boundaries than on interior surfaces. The single-view U-Net obtains an MAE of 1.676 cm in the boundary region and 1.061 cm in the non-boundary region. The nine-view early-fusion U-Net performs slightly worse, with corresponding errors of 1.755 cm and 1.123 cm. The geometry-based multiview model reduces these errors to 0.940 cm and 0.303 cm, respectively. Its boundary error is therefore lower than even the non-boundary error of either U-Net.

The substantially lower error away from object boundaries, with the baseline errors approximately 3.5 to 3.7 times larger, shows that explicit geometric matching considerably improves depth estimation on object surfaces, while the remaining increase at boundaries indicates that depth discontinuities continue to be the more difficult region for every architecture. The early-fusion result again shows that adding source observations without explicitly aligning them does not resolve either type of error.

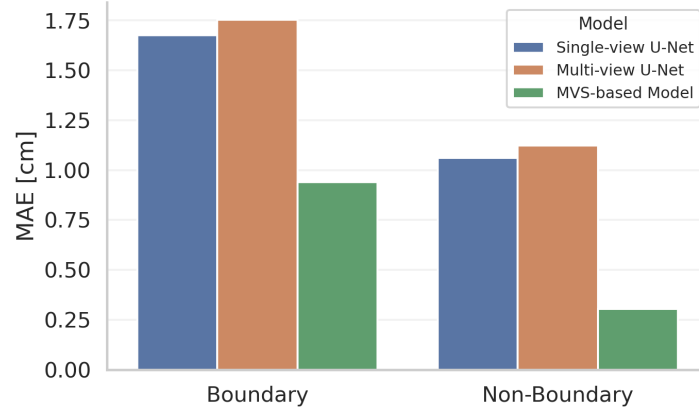


Figure 6.6: MAE in boundary and non-boundary regions of the raised cube for the single-view U-Net, nine-view early-fusion U-Net, and geometry-based multiview model.

6.4.3 Influence of Camera Motion and Event Activity

Event generation depends on changes in image intensity. Consequently, camera motion affects the amount of input signal. Figure 6.7 relates depth error to the translational velocity of the robot end effector, to which the event camera is rigidly attached. Frames are grouped into fixed velocity intervals of 0.01 m s^{-1} . For each populated interval and model, the plotted value is the mean of the per-frame MAE values in that interval. The resulting values are connected by lines to make their variation with velocity easier to follow; the lines do not represent individual frame trajectories. The errors of both U-Net models increase markedly at low velocities, whereas the geometry-based multiview model remains comparatively consistent across the observed velocity range.

A likely explanation is that slow motion produces fewer events and therefore leaves the U-Nets with insufficient evidence for accurate object-depth prediction. This interpretation is supported by Figure 6.8, which relates prediction error more directly to event activity.

While still showing a slight increase in error for low velocities, the effect is much less present for the MVS model than it is for the baseline models. This is likely because the model can compensate for a weak reference observation by aligning evidence from source views that contain stronger or complementary event activity. The nine-view early-fusion model does not show the same robustness, reinforcing the conclusion that access to additional views is not sufficient by itself.



Figure 6.7: Raised cube masked depth error as a function of end-effector translational velocity for the three models. Frames are grouped into fixed 0.01 m s^{-1} velocity intervals, and each plotted value is the mean per-frame MAE within one populated interval. Adjacent interval means are connected by lines to visualize the trend.

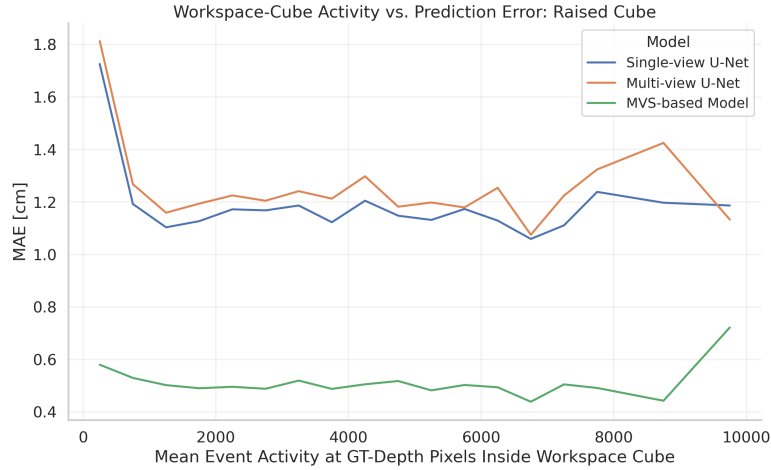


Figure 6.8: Raised cube masked depth error as a function of event activity inside the workspace cube for all three models. Per-frame activity is calculated by summing the absolute event-voxel values over the five temporal bins and over pixels inside the ground-truth workspace-cube mask. Frames are grouped into fixed 500-activity-unit intervals, and each plotted value is the MAE within one populated interval. Adjacent interval means are connected by lines to visualize the trend.

6.4.4 Reconstruction Comparison

The differences observed in the depth-estimation results also carry over to the downstream three-dimensional reconstruction task. To compare the three architectures under identical conditions, the predicted depth maps of the three models are independently integrated into TSDF volumes using the same reconstruction pipeline. The U-Net baselines use uniform TSDF integration, whereas the multiview reconstruction uses its predicted uncertainty for confidence weighting. The resulting meshes are evaluated against the reference reconstructions generated from ground-truth depth. Table 6.8 reports the surface-based reconstruction metrics inside the workspace cube, averaged over the six evaluation objects.

Table 6.8: Surface-based reconstruction metrics inside the workspace cube for all three depth-estimation architectures. Reported values are unweighted means over the six evaluation objects. The U-Net results use uniform TSDF integration and the multiview result uses uncertainty-weighted integration. Lower values are better for accuracy, completeness, and Chamfer distance, whereas higher values are better for normal consistency and F-score.

Model	Accuracy [cm]	Completeness [cm]	Chamfer [cm]	Normal consistency	F@1 cm [%]
Single-view U-Net	0.191	0.175	0.183	0.932	96.18
Nine-view early-fusion U-Net	0.191	0.184	0.187	0.929	95.85
Geometry-based multiview	0.108	0.108	0.108	0.966	99.62

The reconstruction metrics show that the clear advantage of the geometry-based multiview approach in depth estimation also translates to substantially better three-dimensional reconstruction quality. Both U-Net models produce less accurate and, in particular, less complete reconstructions, whereas the geometry-based multiview model reconstructs surfaces that remain close to the reference geometry while covering a substantially larger fraction of the object surface. This difference is reflected by the lower surface-distance errors and the higher normal consistency and F-score of the multiview reconstruction. The qualitative comparison in Figure 6.9 illustrates the same effect for one evaluation object.

The missing regions in the U-Net reconstructions are particularly important because reconstruction requires depth predictions to be consistent across a sequence of viewpoints. Errors that affect only a limited number of pixels in individual depth maps can accumulate during TSDF integration and result in holes or absent surface regions in the final mesh. The stronger completeness of the geometry-based multiview reconstruction therefore indicates that its advantage is not limited to lower per-frame depth error, but also extends to greater consistency of the predicted geometry across viewpoints.

The nine-view early-fusion U-Net does not eliminate these reconstruction failures despite having access to the same number of observations as the

geometry-based multiview model. This is consistent with the depth-estimation results and further indicates that simply providing additional event views is insufficient. Explicitly aligning observations according to the known camera geometry allows the multiview model to combine complementary evidence across viewpoints, resulting in depth predictions that are substantially more suitable for three-dimensional reconstruction.

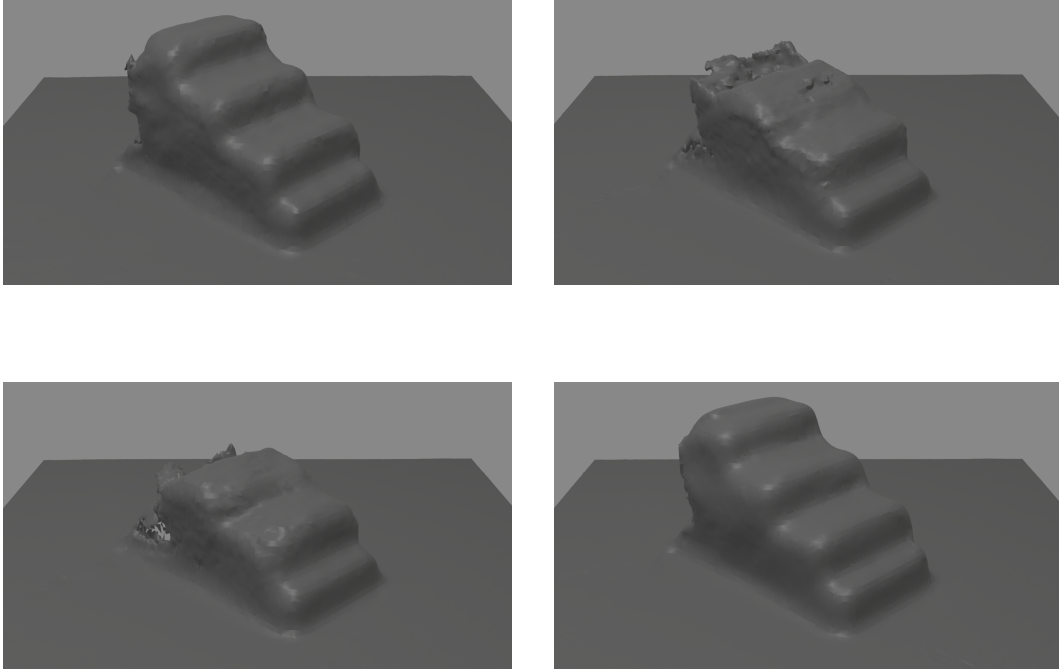


Figure 6.9: Qualitative reconstruction comparison for one evaluation object. Ground-truth reconstruction (upper left), single-view U-Net (upper right), nine-view early-fusion U-Net (lower left), and geometry-based multiview model (lower right). Both U-Net reconstructions contain substantial missing regions, whereas the geometry-based multiview reconstruction recovers a more complete representation of the object surface.

6.4.5 Computational Requirements

The geometry-based multiview model is substantially slower than either U-Net baseline because it constructs and regularizes three cost volumes and performs feature warping for multiple views and depth hypotheses. Nevertheless, its measured throughput of approximately 40 frames per second exceeds the 30 Hz rate of the depth camera used for data collection. The network forward pass is therefore fast enough in principle to process inputs at the sensor rate on the evaluation hardware. Inference times for all models are listed in Table 6.4.5

This measurement does not by itself demonstrate a complete low-latency real-time system. It was obtained with batches of 15 frames and excludes data loading, host-to-device transfer, event voxelization, view selection, and other preprocessing. Moreover, the current view-selection procedure uses four observations acquired after the reference frame. An online implementation would consequently have to buffer future observations before predicting the reference depth map, introducing a motion-dependent delay. A causal system could instead select only current and previous observations, but the effect of that change on accuracy has not been evaluated. Thus, real-time throughput appears feasible, whereas real-time operation with low end-to-end latency would require a dedicated batch-size-one benchmark and a causal input selection pipeline.

Table 6.9: Measured inference throughput on an NVIDIA L40S. Timings were obtained with a batch size of 15 and cover the synchronized forward pass only.

Model	Inference [ms/frame]	Throughput [FPS]
Single-view U-Net	3.94	253.55
Nine-view U-Net	4.35	229.88
Geometry-based multiview	24.93	40.09

Chapter 7

Discussion and Conclusion

This chapter first considers the implications of the depth and reconstruction results, then relates the proposed approach to previous work. The remaining sections define the scope of the conclusions, identify promising directions for further research, and conclude the thesis by summarizing its contributions and answering the central research question.

7.1 Interpretation of Results

The main objective of this thesis is to investigate whether multi-view stereo (MVS) techniques originally developed for RGB images can be adapted to combine multi-view event data for depth prediction and 3D reconstruction. The obtained results provide strong evidence that this is indeed feasible.

While a naive implementation of multiple observations and their corresponding poses is not necessarily useful, implementing them together with an effective geometric mechanism makes accurate prediction possible. The nine-view early-fusion U-Net receives the same number of observations as the proposed multiview model, as well as pose information, but does not improve on the single-view U-Net. Concatenating the views therefore does not, by itself, provide a sufficient inductive bias for learning correspondence and alignment. The geometry-based model instead uses the calibrated poses to warp source features onto explicit depth hypotheses. Its much lower error indicates that this separation between known projective geometry and learned feature matching is well suited to the recorded event data, as quantified across all evaluation regions in Table 6.7.

Despite being clearly outperformed by the geometry-based multiview model, the single-view U-Net achieves surprisingly good depth estimates given that it receives only a single event representation. From a purely geometric perspective, depth cannot in general be uniquely recovered from a single monocular observation. Its performance is therefore likely to rely substantially

on learned priors specific to the controlled dataset. All objects are constructed from similarly sized plastic blocks and are recorded under comparable conditions, allowing the network to exploit cues such as apparent object size, shape, position relative to the supporting plane, and recurring object structure to infer likely depths. The event voxel grid may additionally contribute temporal information. Although treated as a single network input, it aggregates events into multiple temporal bins and therefore retains information about how image structure changes over a short time interval. Such patterns may provide motion-dependent cues that are unavailable in a conventional single RGB image. However, the present experiments do not isolate the contribution of these temporal cues, so their importance cannot be determined conclusively.

The dependence on camera motion and event activity provides another important observation. The U-Net errors increase at low translational speeds and low event activity, whereas the geometry-based multiview model remains more stable. A plausible explanation is that a weak reference observation can be supplemented by source observations containing stronger or complementary events once those observations are aligned geometrically. This interpretation is consistent with both the activity plots and the comparison with the early-fusion model.

The size of the performance difference depends strongly on the evaluated image region. Whole-frame metrics are dominated by the background and the supporting table, both of which are comparatively simple in the present setup. The upper depth limit further reduces the influence of distant background regions, while the table-plane prior provides direct information about the supporting surface. Consequently, similar whole-frame scores do not imply that the models recover the object equally well. The increasing gap inside the workspace cube and, especially, inside the raised object cube shows why the spatially restricted metrics are necessary.

The reconstruction experiment evaluates a property that is not captured by independent per-frame depth metrics: consistency between viewpoints. TSDF fusion can suppress small uncorrelated errors, but systematic or view-dependent errors produce displaced, duplicated, or missing surfaces. The ability to obtain dense reconstructions from the predicted depth maps therefore shows that the outputs are useful beyond the two-dimensional evaluation. The surface-based comparison of all three architectures is reported in Table 6.8.

Finally, the predicted confidence is meaningfully related to depth error and successfully identifies many unreliable estimates. Using this confidence during TSDF integration also improves every reported surface-based reconstruction metric as shown in Table 6.6. The learned confidence therefore does more than correlate with per-pixel error: it provides useful information for suppressing unreliable depth contributions during fusion. The larger relative gains in the raised cube further indicate that confidence weighting is particularly valuable

in the more difficult object region, where prediction errors are more frequent and less dominated by the supporting table. The complete comparison between uniform and confidence-weighted integration is reported in Table 6.6.

7.2 Limitations and Scope

The dataset was deliberately recorded under controlled conditions so that the effect of explicit multiview geometry could be studied without significant changes in scene structure, illumination, or sensor configuration. All objects are constructed from plastic blocks and therefore share related materials, colours, shapes, and geometric primitives. Their configurations nevertheless provide distinct held-out geometries for evaluating generalization within this object family. The resulting conclusions are strongest for this setting. Generalization across object categories and environments remains to be tested. Additionally, because the same held-out validation set is used for checkpoint selection and final evaluation, the reported performance may additionally be subject to some model-selection bias.

The objects are recorded individually on a white, textureless table under similar indoor lighting. This design provides repeatable acquisition conditions and permits a focused comparison of the three architectures. Under camera motion, the table generates relatively few events, while object boundaries often produce a strong response. Foreground and background are therefore frequently distinguishable in the voxel grids. Textured backgrounds, clutter, overlapping or independently moving objects, and large illumination changes would provide a broader and more demanding evaluation.

The approximate supporting-plane prior is a task-specific but practically motivated source of geometric context. Supporting surfaces are common in robotic manipulation, and their location is often already available from robot calibration, workspace definition, or a short geometric initialization. Using this information is especially helpful for an event camera because a static, textureless plane produces little event activity. The prior describes only the known supporting surface. It does not provide the shape or depth of the object that must be reconstructed, and it is supplied consistently to all compared architectures. It should therefore be understood as the use of available workspace knowledge. However, applications without such knowledge would need to estimate such a prior, replace it with more general scene context, or operate without it.

The method also assumes a static scene and accurate camera intrinsics, extrinsics, synchronization, and robot poses. Calibration errors affect both feature warping and the projection of measured depth into the event-camera frame. The RealSense measurements used as supervision can contain missing values, projection artefacts, and occlusion-related discontinuities. These effects

limit the accuracy of the ground truth and are particularly relevant near object boundaries.

The experimental comparison is deliberately narrow. The single-view and nine-view U-Nets isolate the effects of additional observations and explicit geometry, but they do not represent the full range of event-based depth architectures. Existing methods would require adaptation and retraining on the new dataset before a controlled comparison could be made.

7.3 Future Work

The most important outcome of this thesis is the demonstration that central design principles developed for RGB-based multiview stereo can be applied successfully to event voxel grids. This result encourages further investigation of the architecture in more difficult settings and on other event-camera datasets. It also raises a broader question: which other techniques developed for RGB reconstruction and depth estimation remain underexplored for event data, and which of them can be adapted to exploit the complementary temporal information provided by events?

One important next step is therefore to evaluate the method on a broader dataset. Future recordings should include objects with varied materials, curved and irregular surfaces, different scales, textured or cluttered backgrounds, and less controlled illumination. Including multiple objects and independently moving scene elements would test where the static-scene assumption begins to fail. Extending the method to such environments would also require reconsidering the supporting-plane prior used in the present setup. Depending on the scene, this prior could be replaced by a different source of geometric context or removed entirely. In environments with sufficiently textured or event-active background regions, such an explicit prior may be less important because the event data itself can provide more information about the surrounding scene. A larger dataset would also support comparisons with adapted published event-depth methods and provide a stronger basis for measuring out-of-distribution generalization.

7.4 Summary and Conclusion

The work presented in this thesis contributes both a methodological extension of multiview stereo to event data and an experimental framework for evaluating event-based depth estimation and reconstruction. The main contributions are summarized as follows:

1. A synchronized, object-centric dataset for supervised event-based multiview depth estimation. It contains event streams, dense projected depth measurements, and calibrated camera poses for 48 object-specific sequences recorded with a robotic arm.
2. A pose-aware learning-based multiview stereo pipeline that transfers established RGB-based MVS components to event voxel grids. The method combines multiscale feature extraction, differentiable feature warping, variance-based cost-volume construction, three-dimensional regularization, and coarse-to-fine depth estimation.
3. The integration of an approximate supporting-plane prior that gives the network scene-level geometric context in regions with sparse event activation.
4. A task specific evaluation that compares whole-frame, workspace, and raised-cube depth metrics and analyzes performance across sequences, camera motion, and event activity.
5. A dense reconstruction pipeline that combines predicted depth, calibrated poses, confidence estimates, and TSDF fusion, thereby assessing the usefulness and cross-view consistency of the predictions in a robotic three-dimensional reconstruction task.

The experiments answer the central research question by showing that established multiview stereo principles can be transferred to event-based depth estimation. The proposed geometry-based model consistently outperforms the two U-Net baselines, particularly in the object-focused evaluation region, and remains more robust when event activity is low. The comparison with the nine-view early-fusion U-Net indicates that the improvement is not a consequence of simply providing more observations, but of using camera geometry to combine them in a structured way.

The reconstruction experiments further show that this improvement is relevant beyond per-frame depth accuracy. Depth maps predicted by the multiview model can be fused into dense three-dimensional surfaces that closely

match the reference reconstructions. The learned confidence estimates correlate with prediction error and improve all reported surface-based reconstruction metrics when used to weight TSDF integration, with the largest relative gains occurring in the object-focused raised cube.

Overall, the results demonstrate the viability of applying RGB-derived MVS concepts to event voxel grids in a calibrated robotic setup. The conclusions are currently limited to a static, object-centric workspace with an approximately known supporting plane, but they provide a basis for extending the approach to more diverse scenes and acquisition conditions. More broadly, the results suggest that event representations can support learned geometric matching across viewpoints and that further methods from conventional multiview vision may also be adaptable to event-based reconstruction.

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